Faculty of Technology & Engineering

Devang Patel Institute of Advance Technology and Research

Bachelor of Technology Programme

Computer Science & Engineering

ACADEMIC REGULATIONS & SYLLABUS

(Choice Based Credit System)

Academic Year: 2023-2024

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Technology and Engineering

ACADEMIC REGULATIONS

Bachelor of Technology (CSE) Programme

(Choice Based Credit System)

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
Phone: 02697-247500, Fax: 02697-247100, Email: info@charusat.ac.in
www.charusat.ac.in

Year – 2023-2024

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

DEVANG PATEL INSTITUTE OF ADVANCE TECHNOLOGY AND RESEARCH

COMPUTER SCIENCE & ENGINEERING

Vision

To make a leadership presence in Teaching- Learning and Research in the field of Computer Science and Engineering.

Mission

To empower engineers with expertise and exposure to holistic development for solving the challenges of the technological era.

THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1	To establish a strong foundation in students and help them to apply engineering fundamentals, mathematics, science and humanities necessary to formulate, analyze, design and implement engineering solutions.
PEO2	To engage in **research and find the appropriate solutions for the benefits of mankind** through continuous learning, relearning and unlearning.
PEO3	To bring out world class engineers, scientists, entrepreneurs, model citizens, problem solvers and **critical thinkers** with skills developed by practices of advance technologies.
PEO4	To make students aware about their capabilities so that they can analyze real life problems to develop economically viable and socially acceptable solutions that are sustainable using their **technical and analytical** skills.
PEO5	To provide an environment where the students are nurtured in every aspect for their holistic development such that they achieve excellence in professionalism, **interpersonal skills** as well as moral and ethical conduct.
PEO6	To infuse in students, the ability to work as a part of team on **multidisciplinary projects** and develop leadership qualities that meet the challenging and changing global needs.
PEO7	To prepare and make students ready for presenting **themselves on global platform** by encouraging them to adapt rapidly changing industry requirements.

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO1	**Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.

PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO1	The computer science & engineering graduates will be able to analyze, evaluate and device optimize solutions in the areas of algorithms, artificial intelligence, modern networks and data science.
PSO2	Gain expertise in a number of computer science fields and experience an environment conducive to skill building for productive career, entrepreneurship and higher studies.

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes
Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. System of Education

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. Duration of Programme

(i)	Undergraduate programme	(B.Tech)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. Eligibility for admissions

As enacted by Govt. of Gujarat from time to time.

4. Mode of admissions

As enacted by Govt. of Gujarat from time to time.

5. Programme structure and Credits

As per annexure – 1 attached

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7 Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

7.2 Internal Evaluation

As per Annexure – 1 attached

7.3 University Examination

7.3.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.

7.3.2 In order to earn the credit in a course a student has to obtain grade other than FF.

7.4 Performance at Internal & University Examination

7.4.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

7.4.2 A student failing to score 45% of the final examination will get a FF grade.

7.4.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	**≥80**	**≥73 <80**	**≥66 <73**	**≥60 <66**	**≥55 <60**	**≥50 <55**	**≥45 <50**	**<45**
Letter Grade	**AA**	**AB**	**BB**	**BC**	**CC**	**CD**	**DD**	**FF**
Grade Point	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**0**

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i G_i is the Grade Point for the course i and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

(iv) A student will not be allowed to move to third year if he/she has not cleared allthe courses of first year.

(v) A student will not be allowed to move to fourth year if he/she has not clearedall the courses of second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all internal and external evaluation components inevery course; and

9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;

9.14 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA $\geq$ 7.5 & $\leq$10.0
First class:	CGPA$\geq$ 6.0 & < 7.5
Second Class:	CGPA$\geq$ 5.0 & < 6.0

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY (CHARUSAT)

FACULTY OF TECHNOLOGY & ENGINEERING (FTE)

CHOICE BASED CREDIT SYSTEM

FOR

BACHELOR OF TECHNOLOGY & ENGINEERING

A. Choice Based Credit System:

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A 'Core Course' is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An 'Elective Course' is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the
University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE): A 'Programme Elective Course' is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE):

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non Credit Course (NC) - AUDIT Course

A ‘Non Credit Course’ is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student’s Grade Sheet.
Attendance and Course Assessment is compulsory for Non Credit Courses.

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B.TECH PROGRAMME IN CSE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 3	CSE201	JAVA PROGRAMMING	2	4	6	4	30	70	50	50	200
	CSE202	MICROPROCESSOR & COMPUTER ORGANIZATION	3	2	5	4	30	70	25	25	150
	CSE203	DATA STRUCTURES & ALGORITHMS	3	4	7	5	30	70	50	50	200
	CSE204	PROJECT-I		2	2	1			50	50	100
	MA253	DISCRETE MATHEMATICS AND ALGEBRA	4	0	4	4	30	70			100
	XXXXX	UNIVERSITY ELECTIVE – I	2			2	30	70			100
	HS121.02 A	CREATIVITY PROBLEM SOLVING & INNOVATION	0	2	2	2			30	70	100
			12	**16**	**26**	**22**	**150**	**350**	**180**	**220**	**1000**
Sem 4	CSE205	DATA COMMUNICATION & NETWORKING	3	2	5	4	30	70	25	25	150
	CSE206	DATABASE MANAGEMENT SYSTEM	3	4	7	5	30	70	50	50	200
	CSE207	DESIGN ANALYSIS OF ALGORITHM	3	2	5	4	30	70	25	25	150
	CSE208	OPERATING SYSTEM	3	2	5	4	30	70	25	25	150
	CSE209	PROGRAMMING IN PYTHON		2	2	1			25	25	50
	CSE210	PROJECT-II	0	2	2	1			50	50	100
	XXXXX	UNIVERSITY ELECTIVE – II	2			2			30	70	100
	HS111.02 A	HUMAN VALUE & PROFESSIONAL ETHICS	0	2	2	2			30	70	100
			12	**18**	**28**	**23**	**120**	**280**	**235**	**315**	**1000**

List of University Electives			
Code	**University Elective - I (UE - I)**	**Code**	**University Elective - II (UE - II)**
EC281.01	Introduction to MATLAB Programming	EC282.01	Prototyping Electronics with Arduino
CE281.01	Art of Programming	CE282.01	Web Designing
CL281.01	Environmental Sustainability and Climate Change	CL282.01	Basics of Environmental Impact Assessment
EE284	Python Programming	EE288	MAINTENANCE OF HOUSEHOLD APPARATUS
IT283	WEB DESIGNING & UI/UX	IT284	DATA VISULIZATION
ME281.01	Engineering Drawing	ME282.01	Material Science
PH233.01	Fundamentals of Packaging	PH238.02	Cosmetics in daily life
PD260.01	Basic Laboratory Techniques	NR261.01	Life Style Diseases & Management
NR251.01	First Aid & Life Support	PT192.01	Occupational Health & Ergonomics
PT191.01	Health Promotion and Fitness	CA225	Programming the Internet
CA224	Introduction to Web Designing	BM241	Health Care Management
BM231	Banking and Insurance	EE287	MATLAB PROGRAMMING
CL283	SDG HANDPRINT Laboratory		

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CSE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 5	CS350	OPERATING SYSTEM	3	2	5	4	30	70	25	25	150
	CS358	DESIGN & ANALYSIS OF ALGORITHMS	4	2	6	5	30	70	25	25	150
	CS359	SOFTWARE ENGINEERING	3	2	5	4	30	70	25	25	150
	CS348	SOFTWARE GROUP PROJECT-III	0	2	2	1			50	50	100
	CS363	MACHINE LEARNING	4	2	6	5	30	70	25	25	150
	CS343	SUMMER INTERNSHIP-I	0	3	0	3			75	75	150
	CSXXX	PROGRAMME ELECTIVE-II	2	2	4	3	30	70	25	25	150
	HS131.02 A	CREATIVITY PROBLEM SOLVING & INNOVATION	0	2	2	2			30	70	100
			16	**17**	**30**	**27**	**180**	**420**	**180**	**220**	**1100**
Sem 6	CS361	CRYPTOGRAPHY & NETWORK SECURITY	3	2	5	4	30	70	25	25	150
	CS353	THEORY OF COMPUTATION	3		3	3	30	70	-	-	100
	CS362	COMPUTER NETWORKS	3	2	5	4	30	70	25	25	150
	CS364	MODERN ARTIFICIAL INTELLIGENCE	3	2	5	4	30	70	25	25	150
	CS357	SOFTWARE GROUP PROJECT-IV		2	2	1			50	50	100
	CSXXX	PROGRAMME ELECTIVE-II	3	2	5	4	30	70	25	25	150
	HS132.02 A	CONTRIBUTORY PERSONALITY DEVELOPMENT	0	2	2	2			30	70	100
			15	**12**	**27**	**22**	**150**	**350**	**205**	**245**	**900**

CODE	PROGRAMME ELECTIVE-I (PE - I)
CS382	ADVANCED JAVA PROGRAMMING
CS380	MOBILE APPLICATION DEVELOPMENT
CS381	FULL STACK DEVELOPMENT

CODE	PROGRAMME ELECTIVE-II (PE - II)
CS378	VIRTUAL & AUGMENTED REALITY
CS379	WIRELESS COMMUNICATION AND MOBILE COMPUTING
CS383	CYBER SECURITY AND CYBER LAWS

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CSE ENGINEERING

CHOICE BASED CREDIT SYSTEM

Sem	Course Code	Course Title	Teaching Scheme				Examination Scheme				
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 7	CS449	INTERNET OF THINGS	3	2	5	4	30	70	25	25	150
	CS450	DESIGN OF LANGUAGE PROCESSOR	3	2	5	4	30	70	25	25	150
	CS451	ADVANCE COMPUTING	3	2	5	4	30	70	25	25	150
	CS442	DATA SCIENCE & ANALYTICS	3	4	7	5	30	70	100	100	200
	CS452	SOFTWARE GROUP PROJECT - V	0	2	2	1			50	50	100
	CS446	SUMMER INTERNSHIP - II	0	3	0	3			75	75	150
	CSXXX	PROGRAMME ELECTIVE-III	4	2	6	5	30	70	25	25	150
			16	**17**	**30**	**26**	**150**	**350**	**250**	**220**	**1050**
Sem 8	CS453	SOFTWARE PROJECT MAJOR	0	18	18	18			250	350	600
			0	**18**	**18**	**18**			**250**	**350**	**600**

CODE	PROGRAMME ELECTIVE - III
CS473	BLOCKCHAIN TECHNOLOGY & APPLICATIONS
CS474	IMAGE PROCESSING AND COMPUTER VISION
CS475	SOFTWARE DEFINED NETWORK

Note:

- **University Elective (UE): -** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects likeResearch Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc.can be included.

- **Cluster Elective (CT): -** Institutional Electives means common electives amonga cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE): -**
- **Institute Elective (IE):-**
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/programs.

B.Tech. (Computer Science & Engineering) Programme

SYLLABI

(Semester – 3)

CHAROTAR UNIVERSITY OF SCIENCE ANDTECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

MA253: DISCRETE MATHEMATICS AND ALGEBRA

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	0	0	4	4
Marks	100	0	-	100	

Pre-requisite courses:

- Engineering Mathematics

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Predicate Calculus	08
2.	Relations and Lattice	10
3.	Graph Theory	12
4.	Recurrence Relations	05
5.	Abstract Algebra	09
6.	Linear Algebra	16

Total hours (Theory): 60

Detailed Syllabus:

1.	**Predicate Calculus:**	**08 Hours**	**13%**
	Revision: Propositions, connectives, converse, inverse, contrapositive, tautology, contradiction. Logical equivalence. Minimal functionally complete set of connectives. Principle conjunctive normal forms and Principle disjunctive normal forms. Predicate calculus using rules of inferences.		
2.	**Relations and Lattice:**	**10 Hours**	**17%**
	Revision of properties of relations on sets. Representations of relations: graphical and matrix representation. Equivalence relation, covering of a set, partition of a set. Partially ordered sets, totally ordered sets, Hasse diagram. Lattices, sub lattices. Properties of lattices (without proof). Complete lattices, bounded lattices, distributive lattices, complemented lattices and complemented distributive lattices.		
3.	**Graph Theory:**	**12 Hours**	**20%**
	Basic terminologies, Simple graph, Types of graphs. Degree of a vertex, matrix representations of graph. Path and connectivity. Eulerian and Hamiltonian graph. Subgraphs, spanning subgraphs, isomorphic graphs. Planar graphs. Matching in graphs. Graph coloring.		
4.	**Recurrence Relations:**	**05 Hours**	**08%**
	Solutions of recurrence relation by direct methods. Generating functions and solutions of recurrence relation.		
5.	**Abstract Algebra:**	**09 Hours**	**15%**
	Groupoid, semi group, monoid, group. Order of group, order of an element, Lagrange's theorem. Subgroup, cyclic subgroup, permutation group.		
6.	**Linear Algebra:**	**16 Hours**	**27%**
	Vector space: definition and examples. Subspaces. Linear combinations, linearly dependence and linearly independence. Basis and dimension of a vector space. Linear transformations. Null space and range of a linear transformation. Rank - nullity theorem. Isomorphisms.		

Student Learning Outcomes:

At the end of the course the students would be able to

CO1	Develop logical argument using truth table and rules of inferences in predicate calculus.
CO2	Relation and types of relations define on sets and utilize it to construct Hasse diagram and lattices on sets.
CO3	Graph and types of the graphs and identify the real world phenomena in terms of graph theory.
CO4	The concept of recurrence, generating functions and their applications in solving recurrence relations.
CO5	Different algebraic structures like groupoid, semi group, monoid, group , cyclic group and permutation group
CO6	Definition of vector space, concepts of the terms: linear span, linear independence, basis, dimension. Definition and properties of linear transformations, range and kernel of a linear transformation.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	-	1	-	-	-	-	-	-	-	2	-
CO2	3	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	2	1	-	2	-	-	-	-	-	-	-	3	1
CO4	3	-	-	-	1	-	-	-	-	-	-	-	3	-
CO5	3	1	-	-	-	-	-	-	-	-	-	-	2	-
CO6	2	1	-	1	-	-	-	-	-	-	-	-	2	-

Correlation levels 1, 2 or 3 as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Recommended Study Material:

❖ **Text Books:**

1. H. Anton and C. Rorres; Elementary Linear Algebra, Application version, Wiley Edition 2010.
2. Kenneth H. Rosen and Kamala Krithivasan. Discrete mathematics and its applications. Vol. 6. New York: McGraw-Hill, 1995.
3. Tremblay, Jean-Paul, and Rampurkar Manohar. Discrete mathematical structures with applications to computer science. New York: McGraw-Hill, 1975

❖ **Reference Books:**

1. Thomas H. Cormen, E. E. Leiserson, R. L. Rivest and C. Stein, Introduction to algorithms (Vol. 6). Cambridge: MIT press,2001
2. Narsingh Deo, Graph theory with applications to engineering and computer science.

Courier Dover Publications, 2016.

3. B. Kolman and R. C. Busby, Discrete Mathematical Structures for Computer Science, 2nd edition, Prentice-Hall, Englewood Cliffs, New Jersey 1987.
4. Swapan Kumar Sarkar, A Text Book of Discrete Mathematics, S. Chand and Co. New Delhi 2008.
4. D. S. Malik and Mridul K. Sen. Discrete mathematical structures: theory and applications. Course Technology, 2004.
5. D. F. McAllister and D. F. Stanat. Discrete Mathematics in Computer Science. Prentice-Hall, Inc. 1977.

❖ **Web Links:**

Lecture Notes:

1. http://www.cs.yale.edu/homes/aspnes/classes/202/notes.pdf
2. http://home.iitk.ac.in/~arlal/book/mth202.pdf
3. https://web.stanford.edu/class/cs103x/cs103x-notes.pdf
4. https://www.cs.cornell.edu/~rafael/discmath.pdf
5. http://www-sop.inria.fr/members/Frederic.Havet/Cours/matching.pdf
6. http://www-sop.inria.fr/members/Frederic.Havet/Cours/coloration.pdf

Video Lectures:

7. http://www.nptelvideos.in/2012/11/discrete-mathematical-structures.html
8. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-042j-mathematics-for-computer-science-fall-2010/video-lectures/

CSE201: JAVA PROGRAMMING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	4	-	6	**4**
Marks	100	100	-	200	

Pre-requisite courses:

- Basic programming skill.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Fundamental of Programming in Java	02
2.	Class Fundamentals	04
3.	Array & String Handling	02
4.	Inheritance, Interfaces & Packages	04
5.	Exceptions Handling	04
6.	Multithreaded Programming	04
7.	File I/O and NIO	05
8.	Java Collection Frameworks and Generics	05
	Total hours (Theory) :	30
	Total hours (Lab) :	60
	Total hours :	90

Detailed Syllabus:

1.	**Fundamental of Programming in Java**	**02 Hours**	**04%**
	History of Java, Basic overview of java, Bytecode, JVM Buzz-words, Application and applets, Constants, Variables & Data Types, Comment, Operators, Control Flow		
2.	**Class Fundamentals**	**04 Hours**	**09%**
	General form of class, Creating class Overloading methods, Constructor, Declaring Object, Returning objects, using objects as parameters, Assigning object reference variables,		

	Introducing Access control , Understanding static Introducing final, The finalize ( ) method, this keyword ,Garbage collection		
3.	**Array & String Handling**	**02 Hours**	**04%**
	Array basics, String Array, String class, StringBuffer and StringBuilder class, String Tokenizer Class and Object Class		
4.	**Inheritance, Interfaces & Packages**	**04 Hours**	**14%**
	Inheritance: Using super creating multilevel Hierarchy, method overriding, Dynamic method dispatch, abstract classes, Using final with Inheritance, Using Package: Defining package, Finding package and CLASSPATH, Access protection, Importing package, Interface: Defining Interface, Default Methods, Implementing Interface, Variables in Interface		
5.	**Exceptions Handling**	**04 Hours**	**11%**
	Exception types, Try …Catch…Finally, Throw, Throws, creating your own exception subclasses		
6.	**Multithreaded Programming**	**04 Hours**	**16%**
	Life cycle of thread, thread methods, thread priority, thread exceptions, Implementing Runnable interface, Synchronization and Concurrency		
7.	**File NIO**	**05 Hours**	**16%**
	File and Directories, Byte streams and character streams, Random Access Files,NIO: Meta Data File Attributes Buffers, Channels, Recursive Operation.		
8.	**Collection Framework and Generics**	**05 Hours**	**26%**
	Collections of objects, Collections: Sets, Sequence, Map, Understanding Hashing, Use of Array List & Vector, Generics Class, Lamda Expression, Functional Reference, Method Reference, Optional Classes, Processing data with streams		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and implement Object Oriented programming concept using basic syntaxes of control Structures, strings and function for developing skills of logic building activity and garbage collection for saving resources.
CO2	Demonstrate basic problem-solving skills: analyzing problems, modelling a problem as a system of objects, creating algorithms, and implementing models and algorithms in an object-oriented computer language (classes, objects, methods with parameters, abstract classes, interfaces, inheritance and polymorphism).
CO3	Use and develop Concurrency theory: progress guarantees, deadlock, live lock, starvation, linearizability.
CO4	Build and test program using new IO api and exception handling
CO5	Analyze and apply collection framework and generics to solve different data structure algorithms.
CO6	Understand and apply java new features

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	2	3	3	2	-	-	-	-	1	-	-	-	2	-
CO3	2	3	3	3	-	-	-	-	-	-	-	-	2	-
CO4	1	2	3	2	-	-	-	1	1	-	-	-	2	-
CO5	1	2	2	2	-	-	-	-	-	-	-	-	2	-
CO6	1	2	-	-	-	-	-	-	-	-	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- **Text Books:**
  1. Java: The Complete Reference, Ninth Edition by Herbert Schildt , Oracle Press.
  2. Java 8 in Action: Lambdas, Streams, and Functional-style Programming by Alan Mycroft and Mario Fusco, Manning Publication

❖ **Reference Books:**

1. Thinking in Java, Bruce Eckel, Prentice Hall
2. Java: A Beginner's Guide (Sixth Edition) by Herbert Schildt , Oracle Press.
3. Core Java Volume I--Fundamentals (9th Edition) (Core Series) by Cay S. Horstmann, Prentice Hal

❖ **Web Materials:**

1. .https://docs.oracle.com

CSE203: DATA STRUCTURE & ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	**5**
Marks	100	100	-	200	

Pre-requisite courses:

- Programming Language

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Data Structure	04
2.	Linear Data Structure	12
3.	Non-Linear Data Structure	16
4.	Sorting	10
5.	Searching	03
	Total hours (Theory) :	45
	Total hours (Lab) :	60
	Total hours :	105

Detailed Syllabus:

1.	**Introduction**	**04 Hours**	**08%**
	Introduction to data structure (Types of data structure), Introduction to algorithms. Algorithm Analysis and Big O notation, Memory representation of Array: Row Order and Column Order, Abstract Data Types (ADT)		
2.	**Linear Data Structure**	**12 Hours**	**27%**
	Stack: Operations: push, pop, peep, change, Applications of Stack: Recursion: Recursive Function Tracing, Principles of recursion, Tail recursion, Removal of Recursion, Tower of Hanoi, Conversion: Infix to Postfix, Infix to Prefix. Evaluation: Prefix and Postfix expression,		

	Queue Simple Queue: Insert and Delete operation, Circular Queue: Insert and Delete operation, Concepts of: Priority Queue, Double-ended Queue, Applications of Queue, **Linked List:** Memory Representation of LL, Singly Linked List, Doubly Linked List, Circular Linked List, Applications of Linked List		
3.	**Non-Linear Data Structure**	**16 Hours**	**36%**
	Tree: Tree Concepts, Tree Traversal Techniques: Pre-order, Post-order and In-order (Recursive and Iterative), Binary Search Tree: Iterative and Recursive, Balanced Trees (AVL Trees, Applications of Tree, Skip list **Heaps**: Priority queues and Binary Heaps **Graph:** Graph concepts, Memory Representation of Graph, BFS and DFS, Applications of Graph		
4.	**Sorting**	**10 Hours**	**23%**
	Sorting (concepts, Selection Sort, Bubble Sort, Merge Sort, Radix Sort, Insertion Sort, Heap Sort, Quick Sort, Counting sort, Topological sort)		
5.	**Searching**	**03 Hours**	**06%**
	Sequential Search, Binary Search, Hashing, Hashing Functions, Collision-Resolution Techniques, Applications		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand and Implement Algorithms and core Data Structures such as stack, queue, hash table, priority queue, binary search tree and graph in programming language.
CO2	Analyse data structures in storage, retrieval and computation of ordered or unordered data.
CO3	Compare alternative implementations of data structures with respect to demand and performance.
CO4	Describe and evaluate the properties, operations, applications, strengths and weaknesses of different data structures.

CO5	Apply and select the most suitable data structures to solve programming challenges.
CO6	Discover advantages and disadvantages of specific algorithms.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	2	-	-	-	-	-	-	-	-	-	3	-
CO2	-	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	-	3	3	3	-	-	-	-	-	-	-	-	2	-
CO4	-	1	-	1	2	-	-	-	-	-	-	-	2	-
CO5	2	2	2	2	-	-	-	-	-	-	-	2	3	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. An Introduction to Data Structures with Applications, Jean-Paul Tremblay, Paul G. Sorenson, McGraw-Hill.
2. Data structure with C, Lipschutz, TMH
3. Introduction to Algorithms: Cormen, Leiserson, Rivest and Stein: Prentice Hall of India
4. Data Structures and Algorithms: Aho, Hopcroft and Ullmann: Addison Wesley.

❖ **Reference book:**

1. Classic Data structures, D.Samanta, Prentice-Hall International.0
2. Data Structures using C & C++, Ten Baum, Prentice-Hall International.
3. Data Structures: A Pseudo-code approach with C, Gilberg & Forouzan, Thomson Learning.

4. Fundamentals of Data Structures in C++, Ellis Horowitz, Sartaj Sahni, Dinesh Mehta, W. H. Freeman.
5. "A Practical Introduction to Data Structures and Algorithm Analysis" by Clifford A. Shaffer
6. Data Structures and Algorithm in Java: Goodrich and Tamassia: John Wiley and Sons.

- **Web material:**

1. http://www.leda-tutorial.org/en/official/ch02s02s03.html
2. http://www.leda-tutorial.org/en/official/ch02s02s03.html
3. http://www.softpanorama.org/Algorithms/sorting.shtml

CSE202: Microprocessor and Computer Organization

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	0	5	**4**
Marks	100	50	0	150	

Pre-requisite courses:

- Digital Electronics

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Register Transfer and Microoperations	07
2.	Basic Computer Organization and Design	09
3.	Central Processing Unit	04
4.	Pipeline and Vector Processing	02
5.	Computer Arithmetic	06
6.	Memory Organization	05
7.	8086,80186, 80286 Processor, 80386 Processors	10
8.	Current Era of Microprocessors	02
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	90

Detailed Syllabus:

1.	**Register Transfer and Microoperations**	07 Hours	15%
	Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Microoperation, Logic Microoperations, Shift Microoperation, Arithmetic Logic Shift Unit.		
2.	**Basic Computer Organization and Design**	09 Hours	17%
	Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory Reference, Instructions, Input-Output and Interrupt, Complete Computer		

	Description, Design of Basic Computer, Design of Accumulator Logic.		
3.	**Central Processing Unit**	04 Hours	10%
	Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes.		
4.	**Pipeline and Vector Processing**	02 Hours	06%
	Parallel Processing, Pipelining, Arithmetic Pipeline, Instruction Pipeline, RISC Pipeline, Vector Processing, Array Processors.		
5.	**Computer Arithmetic**	06 Hours	15%
	Introduction: Binary, Octal, Decimal, Hexadecimal representation, Integer Numbers: Sign-Magnitude,1's complement,2's complement, Addition and Subtraction, Multiplication Algorithm.		
6.	**Memory Organization**	05 Hours	13%
	Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Virtual Memory.		
7.	**8086, 80186, 80286 Processor, 80386 Processors**	10 Hours	21%
	Architectural differences of 8086, 80186 and 80286 Processors, System Architecture, Registers, Memory management: Segment Translation, Page Translation, Combining Segment and Page Translation.		
8.	**Current Era of Microprocessors**	02 Hours	03%
	Comparison of AMD and Intel Architecture, Features of current era of AMD and Intel processors; Tick-Tock: manufacturing pattern of Intel.		

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Understand and Design of Arithmetic circuit, Logic circuit, Shift circuit and Control Unit circuit by using elements of digital logic.
CO2	Visualize and understand the working of CPU, different instruction formats, addressing modes, pipeline and vector processing. Evaluate the performance of pipeline approach.
CO3	Understand and apply r's and r-1's complement in Addition, Subtraction and Multiplication of signed and unsigned numbers.
CO4	Describe the memory hierarchy and requirements of different memory. Compare and evaluate different cache memory mapping techniques.
CO5	Understand and differentiate processors development from history to current era.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	3	3	-	-	-	-	1	1	-	-	1	2	-
CO2	3	3	3	1	-	-	-	1	1	-	-	-	2	-
CO3	2	2	3	-	-	-	-	1	1	-	-	-	2	-
CO4	3	3	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	3	3	3	-	-	-	-	-	-	-	-	3	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put "-"

Recommended Study Material:

❖ **Text book:**

1. Computer System Architecture, Morris Mano (3rd Edition) Prentice Hall.
2. 80386 Programmer's reference manual from MIT.
3. Microprocessors and Interfacing: Experiments Manual: Programming and Hardware by Douglas V. Hall.

❖ **Reference book:**

1. William Stalling, Computer Organization & Architecture-Designing for Performance, Pearson Prentice Hall (8th Edition).
2. A.S. Tananbum, Structured Computer Organization, Pearson Publisher.
3. The Essentials of Computer Organization and Architecture Linda Null, Julia Lobur.
4. John P Hayes, Computer Architecture & Organization, McGraw-Hill.
5. Computer Architecture: Pipelined and Parallel Processor Design Michael J. Flynn (4th edition).

- **Web material:**
  1. Computer Organization and Architecture: A Pedagogical Aspect: https://onlinecourses.nptel.ac.in/noc21_cs37/course
  2. https://css.csail.mit.edu/6.858/2014/readings/i386.pdf (80386 Programmer Reference Material)

- **Software:**
  1. 8085 &8086 Simulator.
  2. Logisim : http://www.cburch.com/logisim/
  3. Little Man Computer
     a. https://peterhigginson.co.uk/LMC/
     b. https://www.101computing.net/LMC/
     c. https://www.youtube.com/watch?v=fXMCnzdNemc
     d. https://www.youtube.com/watch?v=sEFnRDgkaWA

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE204: PROJECT-I

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	2	-	2	1
Marks	0	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of SoftwareEngineering.
- Students have to submit project with following listed documents at the time of final submission.
  - ❖ Final Project Report
  - ❖ Project Setup file with Source code
  - ❖ Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or projectwork.

Total hours (Theory) : 00

Total hours (Lab) : 30

Total hours : 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Create enhanced employment by moulding the students with higher technical skill.
CO2	Promote creative thinking, to provide hands on actual technology.
CO3	Handle software project and get use to software development processes.
CO4	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Improve communication and presentation skill.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	-	-	-	-	-	-	1	-	-	-
CO2	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO3	-	-	-	-	2	-	-	-	2	-	-	-	-	-
CO4	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	2
CO6	-	-	-	-	-	-	3	1	-	3	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- **Reference book:**
  1. Books, Magazines ,Journals & online course platforms of related topics
- **Web material:**
  1. www.ieeexplore.ieee.org
  2. www.sciencedirect.com
  3. www.elsevier.com
  4. https://www.udemy.com/
  5. https://www.udacity.com/
  6. https://nptel.ac.in/course.html
  7. https://www.futurelearn.com/

- **Software:**
    1. ASP.NET
    2. PYTHON/MATLAB
    3. PHP
    4. ANDROID/IOS

HS121.02 A: CREATIVITY, PROBLEM SOLVING AND INNOVATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	30/15	--	30/15	2
Marks	--	100	--	100	

Pre-requisite courses:

Creative Problem Solving

https://www.coursera.org/learn/creative-problem-solving

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Creativity, Problem Solving and Innovation	06
2.	Questioning, Learning and Visualization	06
3.	Creative Thinking and Problem Solving	06
4.	Logic, Language and Reasoning	06
5.	Contemporary Issues and Practices in Creativity and Problem Solving	06
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	30

Detailed Syllabus:

1.	**Introduction to Creativity, Problem Solving and Innovation**	**6 Hours**	**20%**
	Definitions of Creativity and Innovation, Need for Problem Solving and Innovation, Scope of Creativity in various Domains, Types and Styles of Thinking, Strategies to develop Creativity, Problem Solving and Innovation skills		
2.	**Questioning, Learning and Visualization**	**6 Hours**	**20%**
	Strategy and Methods of Questioning, Asking the Right Questions, Strategy of Learning and its Importance, Sources and Methods of Learning, Purpose and Value of Creativity Education in real life, Visualization strategies - Making thoughts Visible, Mind Mapping and Visualizing Thinking		
3.	**Creative Thinking and Problem Solving**	**6 Hours**	**20%**
	Creative Thinking and its need, Strategy of Thinking Fluency,		

	Generating all Possibilities, SCAMPER Technique, Divergent Vs Convergent Thinking, Lateral Vs Vertical Thinking, Fusion of Ideas for Problem Solving, Applying strategies for Problem Solving		
4.	**Logic, Language and Reasoning**	**6 Hours**	**20%**
	Basic Concepts of Logic, Statement Vs Sentence, Premises Vs Conclusion, Concept of an Argument, Functions of Language: Informative, Expressive and Directive, Inductive Vs Deductive Reasoning, Critical Thinking & Creativity, Moral Reasoning		
5.	**Contemporary Issues and Practices in Creativity and Problem Solving**	**6 Hours**	**20%**
	Cognitive Research Trust Thinking for Creatively Solving Problems, Case Study on Contemporary Issues and Practices in Creativity and Problem Solving		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Demonstrate creativity in their day to day activities and academic output.
CO2	Solve personal, social and professional problems with a positive and an objective mindset.
CO3	Think creatively and work towards problem solving in a strategic way.
CO4	Initiate new and innovative practices in their chosen field of profession.
CO5	Give logical ideas, opinions, and solutions to problems.
CO6	Think critically over the situation and drawing conclusion.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	-	1	1	-	-	-	2	-	1	1	-
CO2	-	3	2	1	-	-	-	1	1	1	-	1	-	1
CO3	-	2	2	1	-	-	-	-	-	-	-	-	-	-
CO4	1	-	1	-	-	-	2	-	-	-	-	-	1	-
CO5	-	1	-	1	-	-	-	-	1	3	-	1	-	1
CO6	-	-	-	2	-	-	-	-	-	-	2	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- **Text book:**
    1. R Keith Sawyer, ZigZag, The Surprising Path to Greater Creativity,Jossy-Bass Publication 2013
    2. Michael Michalko, Crackling Creativity, The Secrets of Creative Genus, Ten Speed Press 2001

❖ **Reference book:**

1. Michael Michalko, Thinker Toys, Second Edition, Random House Publication 2006
2. Edward De Beno, De Beno’s Thinking Course, Revised Edition, Pearson Publication 1994
3. Edward De Beno, Six Thinking Hats, Revised and Update Edition, Penguin Publication 1999
4. Tony Buzan, How to Mind Map,Thorsons Publication 2002
5. Scott Berkum, The Myths of Innovation, Expended and revised edition, Berkun Publication 2010
6. Tom Kelly and David Kelly, Creative confidence: Unleashing the creative Potential within Us all, William Collins Publication 2013
7. Ira Flatow, The all Laughed, Harper Publication 1992
8. Paul Sloane, Des MacHale& M.A. DiSpezio, The Ultimate Lateral & Critical Thinking Puzzle book,Sterling Publication 2002

❖ **Additional Readings:**

1. Keith Sawer, Group Genius, The Creative Power of Collaboration, Basic Books Publication 2007
2. Edward De Beno, Lateral Thinking, Creativity Step by Step, Penguin Publication 1973
3. Nancy Margulies with Nusa Mall, Mapping Inner Space, Crown House Publication 2002
4. Tom Kelly with Jonathan Littman,The Art of Innovation, Profile Publication 2001
5. Roger Von Oech,A Whack on the Side of the Head. Revised edition, Hachette Publication 1998
6. Roger Von Oech, A Kick in the Seat of the Head, William Morrow 1986
7. Jonah Lehrer,Imagine How Creativity Works,Canongate Books Publication 2012
8. James M Higgins, 101 Creative Problem Solving Techniques, New Management Publication1994
9. Soctt G Isaksen, K Brain Doval, Donald J Treffinger,Creative Approach to Problem Solving, Sage Publication 2000
10. Donald J Treffinger, scott G Isaksen, K Brain stead Dorval Creative Problem Solving An Introduction,Prufrock Press 2006
11. H Scott Fogler& Steven E. LeBlance,Strategies for Creative Problem Solving,Prentice Hall Publication 2008
12. Dave Gray, Sunni Brown and James Macanufo, Game Storming, O'reilly Publication 2010.
13. Howard Gardner, Creating minds,Basic Books Publication 1993
14. MihalyCsikzentmihalyi, Creativity–Flow and Psychology of Discovery and Invention,Harper Publication 1996
15. Martin Gerdner,W. H.,Ahal Insight,Freeman Publication 1978
16. Paul Sloane,Test Your Lateral Thinking IQ,Sterling Publication1994
17. Paul Sloane & Des Machale Intriguing, Lateral Thinking Puzzles,Sterling Publication 1996

❖ **Web material:**

1. Internet Search based May TED talks and other sources for videos, slide shares, problems, etc

B.Tech. (Computer Science & Engineering) Programme

SYLLABI

(Semester – 4)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY&ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE 207: DESIGN & ANALYSIS OF ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Computer Programming.

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	To derive time and space complexity of algorithm.	03
2.	Analysis of Algorithm	06
3.	Greedy Algorithm	07
4.	Divide and Conquer Algorithm	07
5.	Dynamic Programming	08
6.	Exploring Graphs	04
7.	Backtracking & Branch & Bound	05
8.	String Matching and Introduction to NP- Completeness	05

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

Detailed Syllabus:

1.	**Basics of Algorithms and Mathematics**	**03 Hours**	**05 %**
1.1	What is an algorithm?		
1.2	Performance Analysis, Model for Analysis - Random Access Machine (RAM), Primitive Operations		
1.3	Time Complexity and Space Complexity		
2.	**Analysis of Algorithm**	**06 Hours**	**14 %**
2.1	The efficiency of algorithm, average and worst case analysis, elementary operation		
2.2	Asymptotic Notation		
2.3	Analyzing control statement		
2.4	Analyzing Algorithm using Barometer		
2.5	Solving recurrence Equation		
2.6	Sorting Algorithm		
3.	**Greedy Algorithm**	**07 Hours**	**16 %**
3.1	General Characteristics of greedy algorithms		
3.2	Problem solving using Greedy algorithm		
3.3	Making change problem		
3.4	Graphs: Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm		
3.5	Graphs: Shortest paths; The Knapsack Problem; Job Scheduling Problem		
4.	**Divide and Conquer Algorithm**	**07 Hours**	**16 %**
4.1	Multiplying large Integers Problem		
4.2	Binary Search		
4.3	Sorting (Merge Sort, Quick Sort)		
4.4	Matrix Multiplication		
4.5	Exponential		
5.	**Dynamic Programming**	**08 Hours**	**18 %**
5.1	Introduction, The Principle of Optimality		
5.2	Problem Solving using Dynamic Programming – Calculating the Binomial Coefficient		
5.3	Making Change Problem		
5.4	Assembly Line-Scheduling		

5.5	Knapsack Problem		
5.6	Shortest Path		
5.7	Matrix Chain Multiplication		
5.8	Longest Common Subsequence		
6.	**Exploring Graphs & Backtracking**	**04 Hours**	**09 %**
6.1	An introduction using graphs and games,		
6.2	Traversing Trees – Preconditioning Depth First Search-Undirected Graph; Directed Graph, Breath First Search, Applications of BFS & DFS		
7.	**Backtracking & Branch & Bound**	**05 Hours**	**12%**
7.1	Backtracking –The Knapsack Problem; The Eight queens problem, General Template		
7.2	Brach and Bound –The Assignment Problem; The Knapsack Problem, The min-max principle		
8.	**String Matching and Introduction to NP-Completeness**	**05 Hours**	**10%**
8.18	The naïve string matching algorithm		
8.2	The Rabin-Karp algorithm		
8.3	The class P and NP Problems		
8.4	Polynomial reduction		
8.5	NP- Completeness Problem		
8.6	NP-Hard problems		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyze the asymptotic performance of algorithms.
CO2	Derive time and space complexity of different sorting algorithms and compare them to choose application specific efficient algorithm.
CO3	Understand and analyze the problem to apply design technique from divide and conquer, dynamic programming, backtracking, branch and bound techniques and understand how the choice of algorithm design methods impact the performance of programs.
CO4	Understand and apply various graph algorithms for finding shorted path and minimum spanning tree.
CO5	Synthesize efficient algorithms in common engineering design situations.
CO6	Understand the notations of P, NP, NP-Complete and NP-Hard.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	2	-	-	-	-	-	-	-	-	-	2	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO4	2	3	3	1	-	-	-	-	-	-	-	-	2	-
CO5	1	-	1	-	-	-	-	-	-	-	-	2	1	1
CO6	3	1	-	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put "-"

Recommended Study Material:

Text Books:

1. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald Rivest and Clifford Stein, MIT Press

Reference Books:

1. Fundamental of Algorithms by Gills Brassard, Paul Bratley, Pentice Hall of India.
2. Fundamental of Computer Algorithms by Ellis Horowitz, Sartazsahni and sanguthevar Rajasekarm, Computer Sci.P.
3. Design & Analysis of Algorithms by P H Dave & H B Dave, Pearson Education.

Web Materials:

1. http://www.stanford.edu/class/cs161/
2. http://www.itl.nist.gov/div897/sqg/dads/
3. http://highered.mcgraw-hill.com/sites/0073523402/

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE208: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Introduction to computer and computer architecture.

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Process Management	04
3.	Inter process Communication	08
4.	Deadlock	06
5.	Memory Management	08
6.	Input Output Management	07
7.	File Systems	06
8.	Unix/Linux File System	04

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

Detailed Syllabus:

1.	**Introduction**	**02 Hours**	**05%**
1.1	What is an OS? Evolution of OS		
1.2	OS Services		
1.3	**Types of OS**		
1.4	Concepts of OS		
1.5	Different Views Of OS		
2.	**Process Management**		
2.1	Process, Process Control Block, Process States,		
2.2	Threads, Types of Threads and Dispatching, Concurrent Threads		
3.	**Inter process Communication**	**08 Hours**	**15%**
3.1	Race Conditions, Critical Section, Co-operating Thread/Mutual Exclusion		
3.2	Hardware Solution, Strict Alternation, Peterson's Solution		
3.3	The Producer Consumer Problem, Semaphores, Event Counters, Monitors		
3.4	Message Passing and Classical IPC Problems: Reader's & Writer Problem, Dinning Philosopher Problem.		
4.	**Deadlock**	**06 Hours**	**15%**
4.1	Deadlock Problem, Deadlock Characterization		
4.2	Deadlock Detection, Deadlock recovery		
4.3	Deadlock Prevention.		
4.4	CPU Scheduling, Protection: Address space, Address Translation		
5.	**Memory Management**		
5.1	Paging: Principle of Operation, Page Allocation, H/W Support For Paging		
5.2	Multiprogramming with Fixed partitions		
5.3	Segmentation		
5.4	Swapping		
5.5	Virtual Memory: Concept, Performance Of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets		
6.	**Input Output Management**	**07 Hours**	**15%**
6.1	I/O Devices, Device Controllers, Direct Memory Access		
6.2	Principles of Input/output S/W: Goals of The I/O S/W, Interrupt Handler, Device Driver, Device Independent		
6.3	I/O Software Disks: RAID levels, Disks Arm Scheduling Algorithm, ErrorHandling.		
7.	**File Systems**	**06 Hours**	**15%**

7.1	File Naming, File Structure, File Types, File Access, File Attributes, File Operations, Memory Mapped Files		
7.2	Directories: Hierarchical Directory System, Pathnames, Directory Operations,		
7.3	File System Implementation, Contiguous Allocation, Linked List Allocation, Linked List Using Index, Inodes		
8.	**Unix/Linux File System**	**04 Hours**	**10%**
8.1	Buffer Cache, Inodes, The system calls: ialloc, ifree, namei, alloc and free		
8.2	Mounting and Unmounting, files systems, Network File systems		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Visualize and understand Operating system functionality and working of OS. Understanding of functionality, services of operating system and differentiate between different types of OS.
CO2	Define thread and process. Visualize how processes and threads are managed by the operating systems. Simulate and analyze various process scheduling algorithms. Explain and analyze different Inter process communication techniques
CO3	Describe deadlock and classify detection, recovery, prevention and avoidance algorithms. Test scenarios to report deadlock
CO4	Compare and evaluate various memory management schemes. Simulate and analyze Memory management algorithms. Identify and describe the role of I/O devices.
CO5	Understand the file systems. Understand the secondary storage and simulate disk scheduling algorithm.
CO6	Understand the basic commands of Linux file systems

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	2	-	-	-	-	-	-	1	-	-
CO2	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO3	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO4	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO5	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO6	3	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial

(High) If there is no correlation, put "-"

Recommended Study Material:

- **Text Books:**

  1. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
  2. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley

- **Reference Books:**

  1. Operating Systems, D.M. Dhamdhare, TMH
  2. Operating Systems Internals and Design Principles , William Stallings , Seventh Edition, Prentice Hall
  3. Unix System Concepts & Applications, Sumitabha Das, TMH
  4. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

CSE205: DATA COMMUNICATION & NETWORKING

Credits and Hours:

Teaching Scheme	**Theory**	**Practical**	**Tutorial**	**Total**	**Credit**
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- N/A

Outline of the Course:

Sr. No.	**Title of the unit**	**Minimum number of hours**
1.	Introduction	4
2.	Physical Layer	4
3.	Data Link Layer	10
4.	Medium Access Sub-layer	6
5.	Network Layer	12
6.	Transport Layer	6
7.	Application Layer	3
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	**Introduction and Basic Concepts**	**04 Hours**	**10%**
	Introduction to Computer Networks, Network Architecture, and OSI reference model, services, network standardization.		
2.	**Physical Layer**	**04 Hours**	**10%**
	Basics of data communication, Guided Transmission Media, Wireless Transmission, Switching		

3.	**Data Link Layer**	**10 Hours**	**20%**
	Design issues, Error detection and correction, Elementary Data Link Protocols, Sliding window protocols, Data Link Protocols inInternet		
4.	**Medium Access Sublayer**	**06 Hours**	**10%**
	Channel Allocation, Multiple Access Protocols, IEEE 802 standardsfor Ethernet LAN, Wireless LANS, Broadband Wireless, Data Link Layer Switching		
5.	**Network Layer**	**12 Hours**	**20%**
	Design issues, Routing algorithms, Congestion control, Internetworking-Fragmentation, Addressing, IPv4, IPv6, ARP,DHCP, ICMP, Network layer in the Internet		
6.	**Transport Layer**	**06 Hours**	**10%**
	Design issues, Transport Service, Elements of Transport Protocol, Internet Transport Protocols – UDP & TCP		
7.	**Application Layer**	**03 Hours**	**10%**
	Domain Name System, Electronic mail, WWW		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand and identify different physical layer transmission fundamentals such as types of signals, transmission, multiplexing, types of medium and modulation.
CO2	Evaluate existing layer-2 networking standards and implementations.
CO3	Evaluate key networking protocols, and their hierarchical relationship in the context of a conceptual model, such as the OSI and TCP/IP framework.
CO4	Understand existing different medium access protocols and evaluate for adoption for future networking.
CO5	Understand and differentiate functionality of existing network routing protocols.
CO6	Measure different network parameter such as Throughput & different types of delays.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	1	-	-	-	-	-	-	-	2	-	-
CO2	1	2	1	1	2	-	-	-	-	-	-	2	1	1

CO3	3	2	2	1	2	-	-	-	-	1	-	2	1	1
CO4	2	3	1	1	1	-	-	-	-	2	-	2	-	-
CO5	3	2	3	1	3	-	-	-	1	1	-	3	1	1
CO6	3	3	3	2	3	-	-	-	2	2	-	3	3	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put “-”

Recommended Study Material:

❖ **Text book:**

1. Data communication & Networking, Behrouz Forouzan, McGraw-Hill

❖ **Reference book:**

1. Data and Computer Communications, William Stallings, Prentice Hall
2. Computer Network, Andrew S. Tanenbaum, Fourth Edition, Prentice Hall

❖ **Web material:**

1. www.wikipedia.org
2. http://www.webopedia.com

❖ **Software:**

1. Wireshark
2. Cisco Packet Tracer

CSE206: DATABASE MANAGEMENT SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	4	-	7	**5**
Marks	100	50	-	150	

Pre-requisite courses:

- Data Structure

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introductory concepts of DBMS	03
2.	Relational Model	05
3.	Entity-Relationship model	07
4.	Formal Relational Query Languages	05
5.	Relational Database Design	07
6.	Transaction & Recovery Management	08
7.	Advanced Transaction Processing	06
8.	Database Security	06
9.	Indexing and Hashing	05
10.	Query Processing & Query Optimization	05
	Total hours (Theory) :	45
	Total hours (Lab) :	60
	Total hours :	105

Detailed Syllabus:

1.	**Introductory concepts of DBMS**	**03 Hours**	**06%**
	Introduction and applications of DBMS, Purpose of database, Data Independence, Database System architecture- levels, Mappings, Database users and DBA		
2.	**Relational Model**	**05 Hours**	**15%**
	Structure of Relational Databases, Database Schema, Schema Diagram, Domains, Relations, Relational Query Languages, Relational Operations		
3.	**Entity-Relationship model**	**07 Hours**	**15%**
	Basic concepts, Design process, Constraints, Keys, Design issues, E-R diagrams, Weak Entity Sets, Extended E-R features- Generalization, Specialization, Aggregation, Reduction to E-R database schema		
4.	**Formal Relational Query Languages**	**05 Hours**	**11%**
	The relational Algebra, The Tuple Relational Calculus, The Domain Relational Calculus		
5.	**Relational Database design**	**07 Hours**	**13%**
	Functional Dependency–definition, Trivial and Non-Trivial FD, Closure of FD set, Closure of attributes, Irreducible set of FD, Normalization – 1NF, 2NF,3NF, Decomposition using FD- Dependency Preservation		
6.	**Transaction & Recovery Management**	**08 Hours**	**15%**
	Transaction concepts, Properties of Transactions, Serializability of transactions, Testing for Serializability, System recovery, Two- Phase Commit protocol, Recovery and Atomicity, Log-based recovery¸ Concurrent executions of transactions and related problems, Locking mechanism, Solution to Concurrency Related Problems, Deadlock, Two-phase locking protocol, Intent locking		
7.	**Advanced Transaction Processing**	**06 Hours**	**10%**
	Transaction-Processing Monitors, TransactionalWorkflows, Main-Memory Databases, Real-TimeTransaction Systems, Long-Duration Transactions		

8.	**Database Security**	**06 Hours**	**10%**
	Views - What are views for? View retrievals, View updates, Snapshots (a digression), Materialized view, Security – Security and Authentication, authorization in SQL, Data		
	encryption, Missing Information - An overview of the 3VL approach		
9.	**Indexing and Hashing**	**05 Hours**	**15%**
	Basic Concepts, Ordered Indices, B+-Tree Index Files, B+-Tree Extensions, Multiple-Key Access, Static Hashing, Dynamic Hashing, Comparison of Ordered Indexing and Hashing, Bitmap Indices, Index Definition in SQL		
10.	**Query Processing & Query Optimization**	**05 Hours**	**10%**
	Overview, Measures of Query Cost, Selection Operation, Sorting, Join, Evaluation of Expressions, Transformation of relational Expressions, Estimating Statistics of expression results, Query Evaluation plans		

Course Outcomes (COs):

At the end of the course, students will be able to

CO1	Apply the concepts of engineering i.e collecting data, organize the data in the systematic form, and arrange the data in a computational way and applying mathematics formation.
CO2	Analyze how data are stored and maintained using data models. Ready to assimilate the concept of data abstraction and design queries using SQL. Identify how data is represented in the relational model and create relations using SQL language
CO3	Identify and evaluate the constructs in the E-R model and issues involved in developing an E-R diagram. Convert an E-R diagram into a relational database schema. Declare and enforce integrity constraints on database using a state-of-art RDBMS.
CO4	Produce aggregate operators to write SQL queries which are not expressible in relational algebra. “More mathematical” notation may apply and also used in research and other venues. Combining these concepts allows production of sophisticated queries.

CO5	Decompose un-normalized tables into normalized compliant tables. Design and implement a normalize database schema for a given problem-domain. Produce strategies to minimize risks of security breaches in a range of network environments and data storage systems. Compute retrieval time and concluding with suitable indexing technique.
CO6	Compare transactions and their properties with (ACID) and without ACID. Apply locking protocol to ensure isolation. Develop logging technique to ensure atomicity and durability. Design a logical view which can be used for analytical tasks. Develop practical experience of the design and implement scalable, secure databases.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	1	2	1	1	-	-	-	1	2	-
CO2	3	3	3	2	3	1	2	-	-	-	-	3	2	1
CO3	3	3	3	3	2	2	1	3	2	1	-	2	3	2
CO4	3	3	1	3	1	1	-	2	2	-	-	2	2	2
CO5	3	3	3	2	3	1	-	2	2	1	-	3	2	2
CO6	3	3	2	2	-	2	1	3	1	-	-	3	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put “-”

Recommended Study Material:

- ❖ **Text book:**
    1. Database System Concepts, Abraham Silberschatz, Henry F. Korth& S. Sudarshan, McGraw Hill.
    2. An introduction to Database Systems, C J Date, Addition-Wesley
- ❖ **Reference book:**
    1. “Fundamentals of Database Systems”, R. Elmasri and S.B. Navathe, the Benjamin / Cumming Pub. Co
    2. SQL,PL/SQL the Programming Language of oracle, Ivan Bayross, BPB Publications

3. Oracle: The Complete Reference, George Koch, Kevin Loney, TMH /oracle press

❖ **Web material:**

1. http://www.sql.org
2. http://www.w3schools.com
3. http://www.sqlcourse.com

❖ **Software:**

1. Oracle 10g
2. SQL Lite
3. Live SQL
4. Firebase
5. Squirrel SQL
6. Postgre SQL

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE209: PROGRAMMING IN PYTHON

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	2	0	2	**1**
Marks	0	50	0	50	

Pre-requisite courses:

- ❑ High level language (C/C++/Java)
- ❑ Web Programming

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1	Basics of Python	02
2	Data Structures: Lists, Tuples, Dictionaries and Strings	04
3	Control structures and Function	04
4	Modules and Scoping Rules	02
5	Exceptions Handling	04
6	Magic Methods, Properties, and Iterators	04
7	Object Oriented Programming	06
8	Regular Expression and File Handling	04
	Total hours (Theory) :	0
	Total hours (Lab) :	30
	Total hours :	30

Detailed Syllabus:

1.	**Basics of Python**	**02 Hours**	**07 %**
	Using the Python Interpreter, Variables, Identifiers and Keywords, Numbers and Expressions		
2.	**Data Structures: List, Tuples, Dictionaries and Strings**	**04 Hours**	**13 %**
	Common Sequence Operations: Indexing, Slicing, Adding Sequences, Multiplication, Membership, Length, Minimum, and Maximum, Using Lists as Stacks, Using Lists as Queues, List Comprehensions, Nested List Comprehensions, the del statement, Tuples and Sequences, Sets, Dictionaries, Comparing Sequences and Other Types, Basic String Operations		
3.	**Control Structures and Functions**	**04 Hours**	**13%**
	Conditional Branching: if Statements, break and continue Statements, and else Clauses on Loops, pass Statements Loops: while Loops, for Loops, Defining Functions, More on Defining Functions: Default Argument Values, Keyword Arguments, Arbitrary Argument Lists, Unpacking Argument Lists, Lambda Expressions, Documentation Strings, Function Annotations		
4.	**Modules and Scoping Rules**	**02 Hours**	**07%**
	Executing modules as scripts, The Module Search Path, "Compiled" Python files, Packages: Importing * From a Package, Intra-package References, Packages in Multiple Directories		
5.	**Exception Handling**	**04 Hours**	**13%**
	Syntax Errors, Exceptions, Handling Exceptions, Raising Exceptions, User-defined Exceptions, Defining Clean-up Actions, Predefined Clean-up Actions		
6.	**Magic Methods, Properties and Iterators**	**04 Hours**	**13%**
	Constructors, Item Access: The Basic Sequence and Mapping Protocol, Properties: The property Function, Static Methods and Class Methods, getattr , setattr , and Friends, Iterators, Generators, Generator Expressions		
7.	**Object Oriented Programming**	**06 Hours**	**20%**

	Python Scopes and Namespaces, Class Definition, Class Objects, Instance Objects, Method Objects, Class and Instance Variables, Inheritance, Multiple Inheritance, Private Variables, Polymorphism, Using Properties to Control Attribute Access, Creating Complete Fully Integrated Data Types		
8.	**Regular Expression and File Handling**	**04 Hours**	**14%**
	What is a regular expression?, Regular expressions with special characters, Regular expressions and raw strings, Extracting matched text from strings, Substituting text with regular expressions, Writing and Reading Binary Data, Writing and Parsing Text Files, Iterating over File Contents, Writing and Parsing XML Files, Random Access Binary Files		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Interpret the fundamental python syntax, semantics and fluent in the use of python control flow statements. Express proficiency in the handling of strings and functions.
CO2	Determine the methods to create and manipulate python programs by utilizing the data structures like lists, dictionaries, tuples and sets.
CO3	Identify the commonly used operations involving file systems and regular expressions.
CO4	Articulate the Object-Oriented Programming concepts such as encapsulation, inheritance and polymorphism as used in Python along with magic methods.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	1	1	3	-	-	-	-	-	-	1	-	1
CO2	3	3	2	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	1	1	3	-	-	-	-	-	-	-	2	-
CO4	3	2	2	2	3	-	-	-	-	-	-	-	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put "-"

Recommended Study Material:

- **Text book:**
    1. Magnus Lie Hetland, "Beginning Python From Novice to Professional",Third

Edition, Apress,2017

2. Nigel George, "Mastering Django: Core" Packt Publishing, 2016

- **Reference book:**
  1. David Beazley, Brian K. Jones, "Python Cookbook", 3rd edition, OREILLY,2016
  2. Brett Slatkin, "Effective Python: 59 Specific Ways to Write Better Python", Novatec, 2016
  3. Allen Downey, "Think Python: How to Think Like a Computer Scientist", Green Tea Press,2015
  4. Mark Lutz "Learning Python", 4th Edition, O'REILLY, 2016
  5. Arun Ravindran, Aidas Bendoraitis, Samuel Dauzon, "Django:Web Development with Python",Packt Publishing, 2016

- **Web material:**
  1. https://www.python.org/
  2. http://www.diveintopython3.net/
  3. https://developer.mozilla.org/en-US/docs/Learn/Server-side/Django
  4. https://www.fullstackpython.com/django.html

- **Software:**
  1. Python IDLE
  2. Anaconda Python
  3. PyCharm

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CSE210: PROJECT-II

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	2	-	2	**1**
Marks	0	100	-	100	

Pre-requisite courses:

- Programming Language, Software Engineering.

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programminglanguage.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of SoftwareEngineering.
- Students have to submit project with following listed documents at the time of final submission.
    a. Final Project Report
    b. Project Setup file with Source code
    c. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) :	00
Total hours (Lab) :	30
Total hours :	30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify problems present in society by surveying variety of domains and convert in project definition.
CO2	Explore new ideas and techniques to solve it. Create, select and apply appropriate techniques, resources, modern engineering and IT tools to solve problem.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply ethical principles and commit to responsibilities and norms of the project.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Apply engineering and management principles to achieve project goal.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	-	-	-	-	-	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	-
CO6	3	2	2	1	2	1	-	-	2	3	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put "-"

Recommended Study Material:

- ❖ **Reference book:**
  1. Books, Magazines ,Journals & online course platforms of related topics

Web material:

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. https://www.udemy.com/
5. https://www.udacity.com/
6. https://nptel.ac.in/scourse.html

HS111.02 A: HUMAN VALUES & PROFESSIONAL ETHICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	--	**02/01**	--	30/15	**2**
Marks	**--**	100	--	**100**	

Pre-requisite courses:

- Ethical Leadership through Giving Voice to Values
  https://www.coursera.org/learn/uva-darden-giving-voice-to-values?skipBrowseRedirect=true

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Values and Ethics	05
2.	Elements and Principles of Values	08
3.	Applied Ethics	08
4.	Value, Ethics & Global Issues	05
5.	Contemporary Issues in Values and Ethics	04
	Total hours (Theory):	--
	Total hours (Practical):	30
	Total hours (Lab) :	--
	Total hours :	**30**

Detailed Syllabus:

1.	**Introduction to Values and Ethics**	**5 Hours**	**17%**
	Need, Relevance and Significance of Values General, Concept and Meaning of Values and Ethics		
2.	**Elements and Principles of Values**	**8 Hours**	**26%**
	Universal & Personal Values, Social, Civic & Democratic Value		
3.	**Applied Ethics**	**8 Hours**	**26%**
	Universal Code of Ethics, Professional Ethics, Organizational Ethics, Ethical Leadership, Domain Specific Ethics		
4.	**Value, Ethics & Global Issues**	**5 Hours**	**17%**
	Cross-Cultural Issues, Role of Ethics & Values in Sustainability		
5.	**Contemporary Issues in Values and Ethics**	**4 Hours**	**14%**
	Case Studies, Presentations, Projects		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand the concepts and mechanics of values and ethics.
CO2	Understand the significance of value and ethical inputs in and get motivated to apply them in their life and profession.
CO3	Understand the significance of value and ethical inputs in and get motivated to apply them in social, global and civic issues.
CO4	Develop their responsibility towards society.
CO5	Comprehend their own core values and adhere to those values at their workplace.
CO6	Practice Ethical Leadership.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	1	3	2	1	-	-	-	-
CO2	-	-	-	-	-	1	-	-	1	-	-	2	-	2
CO3	-	-	-	-	-	3	1	-	-	-	-	-	-	1
CO4	-	-	-	-	-	3	1	1	-	-	-	-	-	-
CO5	-	-	1	1	-	-	-	-	2	-	-	-	-	1
CO6	-	-	-	-	-	-	-	3	-	2	-	-	-	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put “-”

Recommended Study Material:

- **Reference book:**

    1. Human Values and Ethics in Workplace, United Nations Settlement Program,2006. (http://www.unwac.org/new_unwac/pdf/HVWSHE/Human%20Values%20&%20Ethics%20-%20Individual%20Guide.pdf).
    2. Ethics for Everyone, Arthur Dorbin, 2009. (http://arthurdobrin.files.wordpress.com/2008/08/ethics-for-everyone.pdf) .
    3. Values and Ethics for 21st Century, BBVA. (https://www.bbvaopenmind.com/wp- content/uploads/2013/10/Values-and-Ethics-for-the-21st-Century_BBVA.pdf)

- **Web material:**

    • www.ethics.org

B.Tech. (Computer Science & Engineering) Programme

SYLLABI

(Semester –5)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS360: MACHINE LEARNING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	Fundamental concepts and Statistical Learning Techniques	06
2.	Neural Networks	08
3.	Bayesian Learning	06
4.	Supervised and Unsupervised Learning	10
5.	Reinforcement learning	02
6.	Kernel Methods	03
7.	Deep Neural Networks	10

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

Detailed Syllabus:

1	**Fundamental concepts and Statistical Learning Techniques**	**06 hours**	**10%**
	Fundamental concepts: Introduction to Data science, Theory and practices in machine learning, Designing a Learning System, Issues in Machine Learning, Applications of ML, Global Developments of ML, Key challenges to adoption of ML in India. **Statistical Learning Techniques:** Descriptive statistics, Simple Linear Regression, ANOVA, Logistic Regression, Multi Linear regression, Correlation, Moving Average, Random Number Generation, Histogram Smoothing, Sampling, Regularization, Rank Percentile.		
2	**Neural Networks**		
	Neurons and biological motivation. Linear threshold units. Perceptrons: representational limitation and gradient descent training, Perceptron learning rule, Hebbian learning rule, Delta Learning rule, Multilayer networks and Backpropagation Learning Algorithm, Feed Forward, Activation Functioning, Types of Neural Network Architecture.	**08 hours**	**20%**
3	**Bayesian Learning**	**06 hours**	**15%**
	Bayes Theorem, Bayes Theorem and Concept Learning, Maximum Likelihood and Least Squared Error Hypothesis, Maximum likelihood hypothesis for Predicting probabilities, Minimum Description Length Principle, Bayes Optimal Classifier, Gibbs Algorithm, Naïve Bayes Classifier, Bayesian Belief Network, EM Algorithm.		
4	**Supervised and Unsupervised Learning**	**10 hours**	**20%**
	Supervised Learning: Classification, Decision Tree, Naïve Bias, Support Vector Machine,		
	Neural Network, K- Nearest Neighbour. **Unsupervised Learning:** Clustering, Nonparametric Methods, K-means, Hierarchical clustering, Density based clustering		
5	**Reinforcement learning**	**02 hours**	**05%**
	Q Learning, Non deterministic rewards and Actions.		
6	**Kernel Methods**	**03 hours**	**10%**
	Support Vector Machine, Sparse kernel machines, Bias-Variance trade-off. Regularization and model/feature selection, Sampling Methods.		
7	**Deep Neural Networks**	**10 hours**	**20%**

Introduction to Deep Learning, Deep Neural Network, RestrictedBoltzmann machine, Convolution Neural Network, AutoEncoders, Deep Belief Network, Recurrent Neural Network, Transfer learning.

Course Outcome:

At the end of the course, the students will be able to

CO1	Apply basic concepts of Machine Learning and Understanding of standard learning algorithms.
CO2	Analyze mathematical modelling of various Machine Learning algorithms
CO3	Understanding challenges of machine learning like data characteristics, model selection, and model complexity
CO4	Identify strengths and weaknesses of machine learning techniques suitable for a given problem domain and data set.
CO5	Design and implement of various machine learning algorithms in a range of real-world applications.
CO6	Evaluate and interpret the results of learning algorithms.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO2	-	2	-	-	-	-	-	-	-	-	-	-	3	-
CO3	3	-	-	-	-	-	3	-	-	-	-	-	-	-
CO4	3	2	-	-	-	-	-	-	-	-	-	-	2	-
CO5	3	-	2	2	2	-	-	-	-	-	2	-	3	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	3	-

Recommended Study Material:

❖ **Text Books:**

1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. ISBN 0070428077
2. Ethem Alpaydin, "Introduction to Machine Learning", MIT Press, 2004

❖ **Reference Books:**

1. Christopher M. Bishop, "Pattern Recognition and Machine Learning", Springer,2006.
2. Richard O. Duda, Peter E. Hart & David G. Stork, "Pattern Classification. SecondEdition", Wiley & Sons, 2001.
3. Trevor Hastie, Robert Tibshirani and Jerome Friedman, "The elements of

statisticallearning", Springer, 2001.
4. Richard S. Sutton and Andrew G. Barto, "Reinforcement learning: Anintroduction", MIT Press, 1998.

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS350: OPERATING SYSTEM

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Introduction to computer and computer architecture.

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	02
2.	Process Management	04
3.	Inter process Communication	08
4.	Deadlock	06
5.	Memory Management	08
6.	Input Output Management	07
7.	File Systems	06
8.	Unix/Linux File System	04

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 7

Detailed Syllabus:

1.	**Introduction**	**02 Hours**	**05%**
1.1	What is an OS? Evolution of OS		
1.2	OS Services		
1.3	Types of OS		
1.4	Concepts of OS		
1.5	Different Views Of OS		
2.	**Process Management**	**04 Hours**	**08%**
2.1	Process, Process Control Block, Process States,		
2.2	Threads, Types of Threads and Dispatching, Concurrent Threads		
3.	**Inter process Communication**	**08 Hours**	**15%**
3.1	Race Conditions, Critical Section, Co-operating Thread/Mutual Exclusion		
3.2	Hardware Solution, Strict Alternation, Peterson's Solution		
3.3	The Producer Consumer Problem, Semaphores, Event Counters, Monitors		
3.4	Message Passing and Classical IPC Problems: Reader's & Writer Problem, Dinning Philosopher Problem.		
4.	**Deadlock**	**06 Hours**	**15%**
4.1	Deadlock Problem, Deadlock Characterization		
4.2	Deadlock Detection, Deadlock recovery		
4.3	Deadlock avoidance: Banker's algorithm for single & multiple resources		
4.4	Deadlock Prevention.		
4.5	CPU Scheduling, Protection: Address space, Address Translation		
5.	**Memory Management**	**08 Hours**	**18%**
5.1	Paging: Principle of Operation, Page Allocation, H/W Support For Paging		
5.2	Multiprogramming with Fixed partitions		
5.3	Segmentation		
5.4	Swapping		
5.5	Virtual Memory: Concept, Performance Of Demand Paging, Page Replacement Algorithms, Thrashing and Working Sets		
6.	**Input Output Management**	**07 Hours**	**14%**
6.1	I/O Devices, Device Controllers, Direct Memory Access		
6.2	Principles of Input/output S/W: Goals of The I/O S/W, Interrupt Handler, Device Driver, Device Independent		
6.3	I/O Software Disks: RAID levels, Disks Arm Scheduling Algorithm, Error Handling.		
7.	**File Systems**	**06 Hours**	**15%**
7.1	File Naming, File Structure, File Types, File Access, File Attributes, File Operations, Memory Mapped Files		

7.2	Directories: Hierarchical Directory System, Pathnames, Directory Operations,		
7.3	File System Implementation, Contiguous Allocation, Linked List Allocation, Linked List Using Index, Inodes		
8.	**Unix/Linux File System**	**04 Hours**	**10%**
8.1	Buffer Cache, Inodes, The system calls: ialloc, ifree, namei, alloc and free		
8.2	Mounting and Unmounting, files systems, Network File systems		
8.3	EXT file system in Linux		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Visualize and understand Operating system functionality and working of OS. Understanding of functionality, services of operating system and differentiate between different types of OS.
CO2	Define thread and process. Visualize how processes and threads are managed by the operating systems. Simulate and analyze various process scheduling algorithms. Explain and analyze different Inter process communication techniques
CO3	Describe deadlock and classify detection, recovery, prevention and avoidance algorithms. Test scenarios to report deadlock
CO4	Compare and evaluate various memory management schemes. Simulate and analyze Memory management algorithms. Identify and describe the role of I/O devices.
CO5	Understand the file systems. Understand the secondary storage and simulate disk scheduling algorithm.
CO6	Understand the basic commands of Linux file systems

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	2	-	-	-	-	-	-	1	-	-
CO2	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO3	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO4	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO5	3	2	-	-	2	-	-	-	-	-	-	-	1	-
CO6	3	2	-	-	2	-	-	-	-	-	-	1	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put "-"

Recommended Study Material:

- **Text Books:**
  1. Modern Operating Systems -By Andrew S. Tanenbaum, Third Edition PHI
  2. Operating System Concepts Avi Silberschatz, Peter Baer Galvin, Greg Gagne, Ninth Edition, Wiley
- **Reference Books:**
  1. Operating Systems, D.M. Dhamdhare, TMH
  2. Operating Systems Internals and Design Principles , William Stallings , Seventh Edition, Prentice Hall
  3. Unix System Concepts & Applications, Sumitabha Das, TMH
  4. Unix Shell Programming, Yashwant Kanitkar, BPB Publications

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CS358: DESIGN & ANALYSIS OF ALGORITHMS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	5
Marks	100	50	-	150	

Pre-requisite courses:

- Computer Programming.

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	To derive time and space complexity of algorithm.	03
2.	Analysis of Algorithm	06
3.	Greedy Algorithm	07
4.	Divide and Conquer Algorithm	07
5.	Dynamic Programming	08
6.	Exploring Graphs	04
7.	Backtracking & Branch & Bound	05
8.	String Matching and Introduction to NP- Completeness	05

Total hours (Theory): 45

Total hours (Lab): 30

Total hours: 75

Detailed Syllabus:

1.	**Basics of Algorithms and Mathematics**	**03 Hours**	**05 %**
1.1	What is an algorithm?		
1.2	Performance Analysis, Model for Analysis - Random Access Machine (RAM), Primitive Operations		
1.3	Time Complexity and Space Complexity		
2.	**Analysis of Algorithm**	**06 Hours**	**14 %**
2.1	The efficiency of algorithm, average and worst case analysis, elementary operation		
2.2	Asymptotic Notation		
2.3	Analyzing control statement		
2.4	Analyzing Algorithm using Barometer		
2.5	Solving recurrence Equation		
2.6	Sorting Algorithm		
3.	**Greedy Algorithm**	**07 Hours**	**16 %**
3.1	General Characteristics of greedy algorithms		
3.2	Problem solving using Greedy algorithm		
3.3	Making change problem		
3.4	Graphs: Minimum Spanning trees (Kruskal's algorithm, Prim's algorithm		
3.5	Graphs: Shortest paths; The Knapsack Problem; Job Scheduling Problem		
4.	**Divide and Conquer Algorithm**	**07 Hours**	**16 %**
4.1	Multiplying large Integers Problem		
4.2	Binary Search		
4.3	Sorting (Merge Sort, Quick Sort)		
4.4	Matrix Multiplication		
4.5	Exponential		
5.	**Dynamic Programming**	**08 Hours**	**18 %**
5.1	Introduction, The Principle of Optimality		
5.2	Problem Solving using Dynamic Programming – Calculating the Binomial Coefficient		
5.3	Making Change Problem		
5.4	Assembly Line-Scheduling		

5.5	Knapsack Problem		
5.6	Shortest Path		
5.7	Matrix Chain Multiplication		
5.8	Longest Common Subsequence		
6.	**Exploring Graphs & Backtracking**	**04 Hours**	**09 %**
6.1	An introduction using graphs and games,		
6.2	Traversing Trees – Preconditioning Depth First Search-Undirected Graph; Directed Graph, Breath First Search, Applications of BFS & DFS		
7.	**Backtracking & Branch & Bound**	**05 Hours**	**12%**
7.1	Backtracking –The Knapsack Problem; The Eight queens problem, General Template		
7.2	Brach and Bound –The Assignment Problem; The Knapsack Problem, The min-max principle		
8.	**String Matching and Introduction to NP-Completeness**	**05 Hours**	**10%**
8.18	The naïve string matching algorithm		
8.2	The Rabin-Karp algorithm		
8.3	The class P and NP Problems		
8.4	Polynomial reduction		
8.5	NP- Completeness Problem		
8.6	NP-Hard problems		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyze the asymptotic performance of algorithms.
CO2	Derive time and space complexity of different sorting algorithms and compare them to choose application specific efficient algorithm.

CO3	Understand and analyze the problem to apply design technique from divide and conquer, dynamic programming, backtracking, branch and bound techniques and understand how the choice of algorithm design methods impact the performance of programs.
CO4	Understand and apply various graph algorithms for finding shorted path and minimum spanning tree.
CO5	Synthesize efficient algorithms in common engineering design situations.
CO6	Understand the notations of P, NP, NP-Complete and NP-Hard.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	1	-
CO2	2	2	-	-	-	-	-	-	-	-	-	2	2	-
CO3	3	3	3	3	2	-	-	-	-	-	-	2	2	-
CO4	2	3	3	1	-	-	-	-	-	-	-	-	2	-
CO5	1	-	1	-	-	-	-	-	-	-	-	2	1	1
CO6	3	1	-	-	-	-	-	-	-	-	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial

(High)If there is no correlation, put “-”

Recommended Study Material:

Text Books:

1. Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald Rivest and Clifford Stein, MIT Press

Reference Books:

4. Fundamental of Algorithms by Gills Brassard, Paul Bratley, Pentice Hall of India.
5. Fundamental of Computer Algorithms by Ellis Horowitz, Sartazsahni and sanguthevar Rajasekarm, Computer Sci.P.
6. Design & Analysis of Algorithms by P H Dave & H B Dave, Pearson Education.

Web Materials:

4. http://www.stanford.edu/class/cs161/
5. http://www.itl.nist.gov/div897/sqg/dads/
6. http://highered.mcgraw-hill.com/sites/0073523402/

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS359: SOFTWARE ENGINEERING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Introduction to Computer Programming
- Database Management System

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction to Software and Software Engineering	4
2	Agile Development	4
3	Managing Software Project	5
4	Requirement Analysis and Specification	4
5	Software Design	5
6	Software Coding &Testing	6
7	Quality Assurance and Management	5
8	Software Maintenance and Configuration Management	5
9	Introduction to SaaS	3
10	Advanced Topics in Software Engineering	4

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

Detailed Syllabus:

1	**Introduction to Software and Software Engineering**	**04 Hours**	**9%**
1.1	The Evolving Role of Software		
1.2	Software: A Crisis on the Horizon and Software Myths		
1.3	Software Engineering: A Layered Technology		
1.4	Software Process Models, The Linear Sequential Model, The		
	Prototyping Model, The RAD Model, Evolutionary Process		
	Models, Agile Process Model		
1.5	Component-Based Development, Process, Product and Process		
2.	**Agile Development**	**04 Hours**	**9%**
2.1	Agility and Agile Process model		
2.2	Extreme Programming		
2.3	Other process models of Agile Development and Tools		
3	**Managing Software Project**	**05 Hours**	**11%**
3.1	Software Metrics (Process, Product and Project Metrics)		
3.2	Software Project Estimations		
3.3	Software Project Planning (MS Project Tool)		
3.4	Project Scheduling & Tracking		
3.5	Risk Analysis & Management(Risk Identification, Risk		
	Projection, Risk Refinement ,Risk Mitigation)		
4	**Requirement Analysis and Specification**	**04 Hours**	**9%**
4.1	Understanding the Requirement		
4.2	Requirement Modeling		
4.3	Requirement Specification (SRS)		
4.4	Requirement Analysis and Requirement Elicitation		
4.5	Requirement Engineering		
5	**Software Design**	**05 Hours**	**11%**
5.1	Design Concepts and Design Principal		
5.2	Architectural Design		
5.3	Process and Component Level Design (Function Oriented Design, Object Oriented Design-UML) (MS Visio Tool)		
5.4	User Interface Design		

5.5	Web Application Design, Design Patterns		
6.	**Software Coding & Testing**	**06 Hours**	**13%**
6.1	Coding Standard and Coding Guidelines		
6.2	Code Review		
6.3	Software Documentation		
6.4	Testing Strategies		
6.5	Testing Techniques and Test Case, Test Suites Design		
6.6	Testing Conventional Applications		
6.7	Testing Object-Oriented Applications		
6.8	Testing Web and Mobile Applications, Testing Tools (Win		
	runner, Load runner)		
7	**Quality Assurance and Management**	**05 Hours**	**11%**
7.1	Quality Concepts and Software Quality Assurance		
7.2	Software Reviews (Formal Technical Reviews)		
7.3	Software Reliability		
7.4	The Quality Standards: ISO 9000, CMM, Six Sigma for SE.		
7.5	SQA Plan		
8	**Software Maintenance and Configuration Management**	**05 Hours**	**11%**
8.1	Types of Software Maintenance, Re-Engineering, Reverse		
	Engineering, Forward Engineering		
8.1	The SCM Process, Identification of Objects in the Software		
	Configuration		
8.2	Version Control and Change Control		
9.	**Introduction to SaaS and SOA**	**03 Hours**	**7%**
9.1	Product Lifetime: Independent Product Vs. Continues		
	Improvement		
9.2	Software as a Service		
9.3	Service Oriented Architecture		
9.4	Cloud Computing		
9.5	SaaS Architecture		

10	**Advanced Topics in Software Engineering**	**04 Hours**	**9%**
10.1	Component-Based Software Engineering, Client/Server Software Engineering, Web Engineering, Reengineering, Computer-Aided Software Engineering, Real-Time Software Engineering		
10.2	Software Process Improvement		
10.3	Design Thinking		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand basics about software engineering principles, methods and practices and to analyze software requirement specification Prepare, SRS (Software Requirement Specification) document andSPMP (Software Project Management Plan) document.
CO2	Apply the concept of Functional Oriented and Object-Oriented Approach for Software Design, To explain the software design strategies and to apply software measurement and metrics using Functionpoint, Cyclomatic complexity and Heal stead software science measures.
CO3	Recognize how to ensure the quality of software product, different qualitystandards and software review techniques.
CO4	Formulate problem by following Software Testing Life Cycle. Apply various testing techniques andtest plan in. Design Manual Test cases for Software Project. Use automation testing tool students will be able test the software.
CO5	Able to understand modern Agile Development and Service OrientedArchitecture Concept of Industry.
CO6	Analyze software risk with estimation parameters such as cost, effort, schedule/duration and understand the concepts of software maintenance,reengineering, reverse engineering, software configuration management.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	1	-	-	3	-	-	-	-	2	-
CO2	2	2	2	1	2	1	-	-	2	-	1	-	2	-
CO3	1	2	3	2	1	1	1	-	2	1	1	1	-	-
CO4	1	3	2		1	-	-	-	1	-	-	1	2	-
CO5	-	-	-	-	-	1	-	2	1	1	1	1	1	1
CO6	-	2	-	-	-	-	1	-	2	2	1	2	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If

there is no correlation, put “-“

Recommended Study Material:

- **Text Books:**
  1. Roger S. Pressman, Software engineering- A practitioner's Approach, McGraw-Hill International Editions
  2. Ivar Jacobson, Object Oriented Software Engineering: A Use Case DrivenApproach, 1st edition

- **Reference Books:**
  1. Engineering Software as a Service An Agile Software Approach, Armando Foxand David Patterson
  2. Ian Sommerville, Software engineering, Pearson education Asia
  3. Pankaj Jalote, An Integrated Approach to Software Engineering by, Springer
  4. Rajib Mall, Fundamentals of software Engineering, Prentice Hall of India.
  5. John M Nicolas, Project Management for Business, Engineering and Technology,Elsevier

- **Web Materials:**
  1. www.en.wikipedia.org/wiki/Software_engineering
  2. www.win.tue.nl
  3. www.rspa.com/spi
  4. www.onesmartclick.com/engsineering/software-engineering.html
  5. www.sei.cmu.edus
  6. https://www.edx.org/school/uc-berkeleyx

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CS382: ADVANCED JAVA PROGRAMMING (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	-	4	3
Marks	100	50	-	150	

Pre-requisite courses:

- Java Programming.

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of Hours
1.	JDBC Programming	05
2.	Servlet	06
3.	Java Server Pages	05
4.	Java Server Faces 2.0	03
5.	Hibernate4.0	05
6.	JAVA web Frameworks: Spring MVC	06

Total Hours (Theory): 30

Total Hours (Lab): 60

Total Hours: 90

Detailed Syllabus:

1	**JDBC Programming**	**05 Hours**	**16%**
1.1	The JDBC Connectivity Model		
1.2	Database Programming: Connecting to the Database, Creating a		
	SQL Query, Getting the Results, Updating Database Data		
1.3	Error Checking and the SQLException Class, The		
	SQLWarning Class		
1.4	The Statement Interface, The ResultSet Interface, Updatable		
	Result Sets		
1.5	Executing SQL Queries, ResultSetMetaData, Executing SQL		
	Updates, Transaction Management.		
2	**Servlet**	**06 Hours**	**20%**
2.1	Servlet Model: Overview of Servlet, Servlet Life Cycle, HTTP		
	Methods		
2.2	Structure and Deployment descriptor		
2.3	ServletContext and ServletConfig interface, Attributes in		
	Servelt, Request Dispacher interface.		
2.4	The Filter API: Filter, FilterChain, Filter Config.		
2.5	Cookies and Session Management: Understanding state and		
	session, Understanding Session Timeout and Session Tracking,		
	URL Rewriting.		
3	**Java Server Pages**	**05 Hours**	**17%**
3.1	JSP Overview: The Problem with Servlets, Life Cycle of JSP		
	Page, JSP Processing, JSP Application Design with MVC,		
	Setting Up the JSP Environment		
3.2	JSP Directives, JSP Action, JSP Implicit Objects		
3.3	JSP Form Processing, JSP Session and Cookies Handling, JSP		
	Session Tracking		
3.4	JSP Database Access, JSP Standard Tag Libraries, JSP Custom		
	Tag, JSP Expression Language, JSP Exception Handling, JSP XML Processing.		
4	**Java Server Faces2.0**	**03 Hours**	**10%**

4.1	Introduction to JSF, JSF request processing Life cycle, JSF Expression Language		
4.2	JSF Standard Component, JSF Facelets Tag, JSF Convertor		
	Tag, JSF Validation Tag		
4.3	JSF Event Handling and Database Access,		
	JSF Librarie: PrimeFaces		
5	**Hibernate**	**05 Hours**	**17%**
5.1	Overview of Hibernate, Hibernate Architecture		
5.2	Hibernate Mapping Types**,** Hibernate O/R Mapping, and		
	Hibernate Annotation.		
5.3	Hibernate Query Language		
6	**Java Web Frameworks: Spring MVC**	**06 Hours**	**20%**
6.1	Overview of Spring, Spring Architecture, bean life cycle		
6.2	XML Configuration on Spring, Aspect – oriented Spring		
6.3	Managing Database, Managing Transaction		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Illustrate database access and details for managing information using the JDBC API
CO2	Design to map java classes and object associations to relational database tables with hibernate framework.
CO3	Demonstrate web application lifecycle using servlet, and JSP.
CO4	Identify the user interface control and build custom tag library using JSF
CO5	Develop real-world applications using java frameworks and understands about MVC architecture with its application.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	1	2	-	-	-	-	-	-	-	1	-
CO2	3	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	2	2	2	2	3	-	-	-	-	-	-	-	-	-

CO4	1	2	2	1	3	-	-	-	-	-	-	-	1	-
CO5	2	3	3	2	3	-	-	-	-	-	-	-	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If

there is no correlation, put “-”

Recommended Study Material:

Text Books:

1. SCWCD, Matthew Scarpino, Hanumant Deshmukh, Jignesh Malavie, Manning publication
2. Hibernate 2nd edition, Jeff Linwood and Dave Minter, Beginning Aprèspublication
3. Java Persistence with MyBatis 3, K. Siva Prasad Reddy, PACKT publication
4. Spring in Action 3rd edition , Craig walls, Manning Publication
5. Java Server Faces in Action, Kito D. Mann, Manning Publication

Reference Books:

1. JDBC™ API Tutorial and Reference, Third Edition, Maydene Fisher, Jon Ellis, Jonathan Bruce, Addison Wesley
2. Beginning JSP, JSF and Tomcat, Giulio Zambon, Apress
3. JSF2.0 CookBook, Anghel Leonard, PACKT publication

Web Materials:

1. http://www.servicearchitecture.com/applicationservers/articles/j2ee_web_site_architecture.html
2. http://www.oracle.com/technetwork/java/javaee/overview/index.html
3. http://www.roseindia.net/struts/hibernate-spring/index.shtml
4. http://www.roseindia.net/jsf/

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY&ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS380: MOBILE APPLICATION DEVELOPMENTS (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	2	2	4	3
Marks	100	50	150	

Pre-requisite courses:

- Basic Design concept with XML, Database management system.

Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1.	Getting an Overview of Android	2
2.	Working with the User Interface Using Views and View Groups	6
3.	Intents and Fragments in Android	4
4.	Database Connectivity	3
5.	Introduction to Xcode and InterfaceBuilder for iOS	3
6.	Model Development with Swift	6
7.	Intro to Scrollable Views, Tabs and Pages	3
8.	Displaying and Persisting Data	3

Total hours (Theory): 30Hrs.

Total hours (Lab): 60 Hrs.

Total hours: 90 Hrs.

Detailed Syllabus:

1.	**Getting an Overview of Android** Android OS Architecture, Introducing Development Framework,	**02 Hours**	**08 %**
	Dalvik Virtual Machine – DVM, Android Virtual Device and SDK		
	Manager, Developing and Executing the First Android Application,		
	Android Activities- Creating an Activity, Managing the Lifecycle of		
2.	an Activity, **Working with the User Interface Using Views and ViewGroups**	**06**	**18**
	Working with Views- Text, EditText, Button, Radio Button,		
	CheckBox, ImageButton, ToggleButton, RatingBar, Working with		
	View Groups- LinearLayout, RelativeLayout, ConstraintLayout,		
	ScrollView, Table, Frame, Table with ActionBar, Binding Data with		
	the AdapterView Class- ListView, Spinner, GallaryView, Creating		
	Menus & Dialogs		
3.	**Intents and Fragments in Android** Intent Objects, Intent Filters, Linking the Activities Using Intent,	**04 Hours**	**14 %**
	Obtaining Results from Intent, Passing Data Using an Intent Object,		
	Fragments- Fragment Implementation, Finding Fragments, Adding,		
	Removing, and Replacing Fragments		
4.	**Database Connectivity**	**03**	**09**
	SQLite Database, SQLite Data Types, Cursors and Content Values,		
	SQLite Open Helper, Adding, Updating and Deleting Content, XML		
	& JSON Based Web Services, Firebase for Android, Firebase		
	connectivity		
5.	**Introduction to Xcode and InterfaceBuilder for iOS** Xcode Intro: Demo of a basic iOS App, StoryBoards, Source files &	**03 Hours**	**09 %**
	wiring them together, COCOA and MVC Framework, Overview of		
	features of latest iOS.		
6	**Model Development with Swift** Swift language essentials: Data types, variables, constants,	**06 Hours**	**18 %**
	operators, Decision making statements, looping, arrays, dictionaries,		
	functions, enumerations, structure, classes, inheritance, Simple		
7	connections to the User Interface **Introduction to Scrollable Views, Tabs and Pages**	**03**	**14**
	Frames and Bounds, Auto Layout, Views, Outlets and Actions,		
	Different View Controller: single view Controller, Master-Detail View Controller, Navigation View Controller, UI Controllers: Label,		

	Button, Text Field, Slider, Switch, Progress View, Page Control.		
8	**Displaying and Persisting Data** Using the Table View, ScrollViews, Collection View, Image View,	**03 Hours**	**10%**
	Text View, Web View, Map View, Date Picker. JSON parsing, XML		
	Parsing in iOS.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand various technologies and business trends impacting mobile applications
CO2	Apply a deep knowledge of mobile device, features, architecture and android functionality.
CO3	Analyse and implement frameworks, database and design patterns in Mobile Applications
CO4	Create a small but realistic working mobile application using features such as data persistence and data communications
CO5	Create a mobile application using the Swift programming language.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	-	1	-	-	-	-	-	-	2	1	-
CO2	3	1	3	2	2	-	-	-	-	-	-	2	2	1
CO3	2	3	3	1	2	-	-	-	1	1	-	2	1	-
CO4	3	2	2	3	2	-	-	-	2	2	-	3	2	1
CO5	2	1	2	1	3	-	-	-	1	-	-	2	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3:

Substantial (High)If there is no correlation, put "-"

Recommended Study Material:

- **Text book:**
  1. Android Developer Tools Essentials by Mike Wolfson - O'Reilly Media Publications
  2. Christian Keur and Aaron Hillegass, iOS Programming: The Big Nerd Ranch Guide, 5thedition, 2015
- **Reference book:**
  1. Learn Java for Android Development, 2nd Edition - Jeff Friesen - Apress Publications

2. Suzanne Ginsburg, Designing the iPhone User Experience: A User-Centered Approachto Sketching and Prototyping iPhone Apps, Addison-Wesley Professional, 2010
3. Bill Phillips, Chris Stewart, Brian Hardy, and Kristin Marsicano, Android Programming:The Big Nerd Ranch Guide, Big Nerd Ranch LLC, 2nd edition, 2015.

- **Web material:**
  1. http://www.youtube.com/watch?v=SUOWNXGRc6g&list=PL2F07DBCDCC01493A
  2. Study Tutorial: https://developer.android.com/sdk/index.html
  3. https://www.xamarin.com/forms
  4. https://docs.microsoft.com/en-us/xamarin/
  5. https://developer.apple.com/xcode/
- **Software:**
  1. Android Studio
  2. Flutter
  3. Xcode

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY&ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS381: FULL STACK DEVELOPMENT (PE-I)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	2	2	-	4	3
Marks	100	50	-	150	

Pre-requisite courses:

- HTML
- CSS
- JAVASCRIPT

Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1.	Basics of Web Development with HTML, CSS and JavaScript	04
2.	Introduction to UI and UX in Web Development	02
3.	Front-end Web Development with React.js	06
4.	React Forms, Flow Architecture and Introduction to Redux	04
5.	Client-Server Communication and React Rendering	04
6.	Server-side Development with Node.js, Express and MongoDB	06
7.	Deployment of MERN Stack Web Applications	04

Total Hours (Theory): 30

Total Hours (Lab): 64

Total Hours: 94

Detailed Syllabus:

1.	**Basics of Web Development with HTML, CSS and JavaScript.**	**04 Hours**	**12%**
	Fundamentals of HTML, CSS and JavaScript, Pseudo classes, Pseudo-elements, Transitions, Animations, Positioning, Box Model, Bootstrap 4, Material-UI, Basics of Responsive website using media queries, DOM Manipulation, Handling Events using JavaScript, Regular Expressions		
2.	**Introduction to UI and UX in Web Development**	**02 Hours**	**8%**
	Visual elements of User Interface Design, Ideation, Articulation and development in UX Design, Look and Feel/Visual Research, Developing and Refining UI, Sitemap, Wireframes and Prototypes		
3.	**Front-end Web Development with React.js**	**06 Hours**	**20%**
	Introduction to Frontend web development, Single page web applications, ECMAScript 6, Fundamentals of React.js, Components: Class Components and Functional Components, Refs, Refs with class components and Forwarding Refs, React Hooks - useEffect, useReducer, useContext, useCallback, fetching data with React Hooks, JSX, Props, Methods as props, State and setState, Destructuring props and state, Event Handling, Rendering, Error Boundary.		
4.	**React Forms, Flow Architecture and Redux**	**04 Hours**	**14%**
	Basics of Form Handling, Controlled Forms and form Validation, Uncontrolled Components and forms, The Model-View-Controller Framework, The Flux Architecture, Introduction to Redux, React Redux Forms, React Redux Form Validation		
5.	**Client-Server Communication and React Rendering**	**04 Hours**	**14%**
	Client-Server communication, HTTP Get Request, HTTP Post Request, React Render: Rendering, useState, state immutability, parent and child components, State Management		
6.	**Server-side Development with Node.js, Express and MongoDB**	**06 Hours**	**20%**
	Introduction to Server-side Development: Node, Node modules, Node HTTP server, Application Programming Interface, Express Framework and Rest API using Express, HTTPS and Secure Communication, Cross-Origin Resource Sharing, Basic Authentication, Session Based Authentication and Token Based Authentication, JSON Web Tokens,		

	Passport module, OAuth and User Authentication, Backend as a Service, Integrating the React Client and Server, MongoDB and Mongoose		
7.	**Deployment of MERN Stack Web Applications**	**04 Hours**	**12%**
	Fundamentals of Secure Deployment, Search Engine Optimization, DNS Record Management, Deploying MERN Stack Application onWeb, Free Hosting Platforms, GitHub, Firebase, Heroku		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand various technologies and trends impacting single page web applications.
CO2	Apply a deep knowledge of MVC (Model View Controller) architecture, making the development process easier and faster using open-source technologies.
CO3	Implement UI-UX using React.js framework and Responsive web designing for Web Applications.
CO4	Demonstrate the use of JavaScript to fulfil the essentials of front-end development to back-end development.
CO5	Understand the use of MongoDB as an open-source NoSQL database to store and retrieve information.
CO6	Create a Real-time responsive web application using MERN STACK technologies.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	2	-	1	-	-	-	-	-	-	2	1	-
CO2	3	1	3	2	2	3	-	-	-	-	-	2	2	1
CO3	2	3	3	1	2	2	-	-	1	1	-	2	1	-
CO4	3	2	2	3	2	-	-	-	2	2	-	3	2	1
CO5	2	1	2	1	3	-	-	-	1	-	-	2	2	-
CO6	3	1	3	2	2	2	-	-	-	-	-	2	2	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If

there is no correlation, put “-”

Recommended Study Material:

- **Textbooks:**
  1. Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node - O'Reilly Media Publications
  2. The Full Stack Developer by Chris Northwood- Apress Publications
- **Reference Books:**
  1. Beginning MERN Stack: Build and Deploy a Full Stack MongoDB, Express, React, Node.js App by Greg Lim
  2. Full-Stack React Projects: Learn MERN stack development by building modern web apps using MongoDB, Express, React, and Node.js, 2nd Edition by Shama Hoque.
- **Prerequisite:**
  1. **HTML**: https://www.w3schools.com/html/
  2. **CSS**: https://www.w3schools.com/css/default.asp
  3. **JavaScript**: https://www.w3schools.com/js/default.asp
  4. **Crash Course Link**:
     https://youtu.be/UB1O30fR-EE,
     https://developer.mozilla.org/en-US/
- **Web Materials:**
  1. h t tps:/ / www.y ou t u b e. com / wat ch ? v=QF aFI cGh P oM& l ist =PLC 3y 8 - r F Hv wg g3v a YJgH Gn M odB54 r xOk3
  2. h t tps:/ / www.y ou t u b e. com / wat ch ? v=7 CqJlx B Yj - M

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY&ENGINEERING

DEPARTMENT OFCOMPUTER SCIENCE & ENGINEERING

CS348: SOFTWARE GROUP PROJECT -III

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	0	**2**	-	2	1
Marks	0	100	-	100	

Outline of theCourses:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most ***three***) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.

  a. Project Synopsis

  b. Software Requirement Specificationc. SPMP

  d. Final Project Report

  e. Project Setup file with Source

  code f. Project Presentation

  (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory): 00

Total hours (Lab): 30

Total hours: 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	An ability to function effectively in teams to accomplish a common goal.
CO2	An ability to apply knowledge of computing and engineering to evaluate project equirements.
CO3	An ability to design, implement and evaluate a computer-based system, process, component, or program to meet desired needs.
CO4	An ability to analyze the local and global impact of computing on individuals, organizations, and society.
CO5	Write technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	An ability to communicate effectively with end user.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	3	2	-	1	2	-
CO2	3	-	1	2	-	-	-	-	-	-	-	-	3	-
CO3	-	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	-	-	-	-	-	3	2	-	-	1	-	-	2	-
CO5	-	-	-	-	-	-	1	-	-	2	-	3	-	2
CO6	-	-	-	-	-	-	3	1	-	-	-	2	-	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic

- **Web Materials:**
    1. www.ieeexplore.ieee.org
    2. www.sciencedirect.com
    3. www.elsevier.com
    4. http://spie.org/x576.xm

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY&ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS343: SUMMER INTERNSHIP-I

Credits and Hours:

Teaching Scheme	Theory	Practical	Project	Total	Credit
Hours/week	0	**0**	-	90	3
Marks	0	0	-	150	

Instruction Method and Pedagogy:

- Summer internship shall be at least 90 hours during the summer vacation only.
- Department/Institute will help students to find an appropriate company/industry/organization for the summer internship.
- The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline.
- Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed.
- During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and also those of the University.
- Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.
- The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
    1. Summer Internship Report
    2. Summer Internship Assessment Form
- Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted to the Coordinator by the first day of the new term.
- The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this

guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.

- If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

Format of Summer Internship Report

The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections:

- Cover Page
- Page of Approval and Grading
- Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at.
- TABLE OF CONTENTS (with the corresponding page numbers)
- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc.

  Provide information regarding

    - Location and spread of the company
    - Number of employees, engineers, technicians, administrators in the company
    - Divisions of the company
    - Your group and division
    - Administrative tree (if available)
    - Main functions of the company
    - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the

summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.

- CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and documentnumber. In this manner you don't have to print it and waste tons of paper.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) If there is no correlation, put “-”

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS131.02 A: COMMUNICATION AND SOFT SKILLS
(HSS ELECTIVE-III)

Credits and Schemes

Sem	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
V	HS131.02 A	Professional Communicatio	02	02	--	--	30	70	100

Course Outline

Module No.	Title/Topic	Classroom Contact Sessions
1	**An Introduction Professional Communication** - *Concept &Applications of Professional Communication* - *Principles of Professional Communication* - *Communication networks: personal sanctum, professional sanctum, inner circle, and outer circle; managing the networks* - *Communication strategies: communicator, audience, message, channel choice, culture*	**04**
2	**Professional Communication and Rhetorics** - *Concepts of "Communication" and "Professional Communication" and "Rhetorics"* - *Orientation towards the Concepts of Professional Communication and Rhetorics (Speaking)* - *Importance of Ethos, Logos, and Pathos in Professional Communication* - *Principles of Professional Communication (visual, oral and non-verbal)*	**04**

3	**Cross-cultural Communication and Globalization** *Basic Concepts: Culture, Globalization and Cross-cultural Communication* *Communicating with People of Different Cultures* *Conflicts in Cross-cultural Communication and Tactics / techniques to resolve them*	**04**
4	**Written Professional Communication** *Importance of Written Professional Communication* *Letter Writing, E-mail Writing, Report Writing* *Resume Building*	**06**
5	**Academic Writing** *Importance of Academic Writing* *Research Paper Writing* *Article/ Review Writing* *Reference and Citation*	**06**
6	**Effective Presentation Strategies** *Why and How in Presentation* *Audience Analysis and Supporting Material* *Presentation Mechanics and Presentation Process* *Managing Yourself during Q and A Session* *Fundamentals of Persuasions*	**06**
	Total	**30**

Pedagogy

Teaching will be facilitated by reading material, discussions, task-based learning, projects, assignments and interpersonal activities like group work, independent and collaborative study projects and presentations, etc.

Evaluation

Internal Evaluation

Students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Assignment	02	05	10
2	Project	01	15	15
3			Attendance	05
			Total	0

External Evaluation

University Practical Examination will be for 70 marks to be conducted at the end of the semester. Details are:

Sl. No.	Component	Number	Marks per incidence	Tota
1	Practical / Viva	01	70	
			Total	0

Course Outcome (COs):

After completion of the course, the student would:

CO1	Gain basic conceptual understanding of communication skills in Professional settings
CO2	Develop awareness and competence in communicating across cultures in professional settings
CO3	Develop confidence and competence in speaking in formal interviews and to make formal presentations
CO4	Develop necessary soft skills to work collaboratively in a group and participate in a group Discussion for employment and in the workplace.
CO5	Develop team and group dynamics to work well in multi-disciplinary and cross-cultural work environment
CO6	Develop writing skills to prepare CV's, Resumes, Statement of Purpose and preparation of other formal documents.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO2	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO3	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	2	3	-	-	-	2
CO6	-	-	-	-	-	-	-	-	-	3	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books

- Eckhouse, B. E. (1999). *Competitive Communication: A Rhetoric for Modern Business.* Oxford University Press. (for Module - II)
- Koneru, A. (2008). *Professional Communication.* New Delhi: Tata Mcgraw-Hill. (for ModuleI & III)
- Meenakshi Raman, P. S. (2006). *Business Communication.* Meenakshi Raman, Prakash Singh. (for module I and IV)
- Parul Popat, K. K. (2015). *Communication Skills.* New Delhi: Pearson. (for Module V & VI)
- Sanjay Kumar, P. L. (2015). *Communication Skills.* Oxford University Press India. (for Module I, III and V)

B.Tech. (Computer Science & Engineering) Programme

SYLLABI
(Semester – 6)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS361: CRYPTOGRAPHY & NETWORK SECURITY

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Outline of the Course:

Sr No.	Title of the unit	Minimum number of hours
1.	Classical Cryptography	**04**
2.	Shannon's Theory	**03**
3.	Block Ciphers And The Advanced Encryption Standard	**05**
4.	Cryptographic Hash Functions	**05**
5.	The RSA Cryptosystem And Factoring Integers	**05**
6.	Public-Key Cryptography And Discrete Logarithms	**05**
7.	Signature Schemes	**05**
8.	Pseudo-Random Number Generation	**03**
9.	Key Distribution & Key Agreement Schemes	**05**
10.	Public-Key Infrastructure & Multicast Security	**05**

Total hours (Theory): 60
Total hours (Lab): 30
Total hours: 90

Detailed Syllabus:

1.	**Classical Cryptography**	**04 hours**	**09 %**
1.1	Introduction: Some Simple Cryptosystems		
1.2	Cryptanalysis		
2.	**Shannon's Theory**	**03 hours**	**07 %**
2.1	Introduction		
2.2	Elementary Probability Theory		
2.3	Perfect Secrecy		
2.4	Entropy		
2.5	Product Cryptosystems		
3.	**Block Ciphers and The Advanced Encryption Standard**	**05 hours**	**11%**
3.1	Introduction		
3.2	Substitution-Permutation Networks		
3.3	The Data Encryption Standard		
3.4	The Advanced Encryption Standard		
3.5	Modes of Operation		
4.	**Cryptographic Hash Functions**	**05 hours**	**11%**
4.1	Hash Functions and Data Integrity		
4.2	Security of Hash Functions		
4.3	Iterated Hash Functions		
4.4	Message Authentication Codes		
5.	**The RSA Cryptosystem and Factoring Integers**	**05 hours**	**11%**
5.1	Introduction to Public-key Cryptography		
5.2	The RSA Cryptosystem		
5.3	Primality Testing		
5.4	Factoring Algorithms		
5.5	Other Attacks on RSA		
6.	**Public-Key Cryptography and Discrete Logarithms**	**05 hours**	**11%**
6.1	The ElGamal Cryptosystem		
6.2	Algorithms for the Discrete Logarithm Problem		
6.3	Lower Bounds on the Complexity of Generic Algorithms		
6.4	Finite Fields		
6.5	Elliptic Curves		

7.	**Signature Schemes**	**05 hours**	**11%**
7.1	Introduction		
7.2	Security Requirements for Signature Schemes		
7.3	The ElGamal Signature Scheme		
7.4	Variants of the ElGamal Signature Scheme		
8.	**Pseudo-Random Number Generation**	**03 hours**	**07%**
8.1	Introduction and Examples		
8.2	The Blum-Blum-Shub Generator		
8.3	Probabilistic Encryption		
9.	**Key Distribution & Key Agreement Schemes**	**05 hours**	**11%**
9.1	Introduction		
9.2	Diffie-Hellman Key Predistribution		
9.3	Key Distribution Patterns		
9.4	Session Key Distribution Schemes		
9.5	Diffie-Hellman Key Agreement		
9.6	Key Agreement Using Self-Certifying Keys		
9.7	Encrypted Key Exchange		
10.	**Public-Key Infrastructure&Multicast Security**	**05 hours**	**11%**
10.1	Introduction: What is a PKI?		
10.2	Certificates		
10.3	The Future of PKI?		
10.4	Identity-Based Cryptography		
10.5	Introduction to Multicast Security		
10.6	Broadcast Encryption		
10.7	Multicast Re-Keying		

Course Outcome:

After completion of the course students will be able to

CO1	Define various security goal and understand the security policies such as the CIA triad of Confidentiality, Integrity and Availability.
CO2	Classify various forms of security attacks, where they arise, and appropriate tools or mechanism to quantify them.
CO3	Illustrate a basic understanding of cryptography, how it has evolved, and evaluate symmetric key encryption techniques used today.
CO4	Distinguish modern symmetric encryption standard, key distribution scenario and analyze effectiveness in todays' environment.
CO5	Evaluate Asymmetric key encryption techniques, key distribution scenario and calculate public and private components of asymmetric key encryption techniques.
CO6	Develop message integrity and message authentication of message digest.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	-	-	3	1	-	2	1	-	-	-	-	-
CO2	2	3	3	2	3	1	-	-	-	-	-	-	1	-
CO3	2	2	3	2	3	2	-	-	2	1	-	-	2	-
CO4	2	2	3	2	3	-	2	-	2	1	-	-	3	1
CO5	3	2	3	2	1	-	2	1	1	-	-	-	3	1
CO6	2	2	3	1	-	-	2	2	1	-	2	-	2	2

Recommended Study Material:

❖ **Text Books:**

1. Douglas R. Stinson, Cryptography: Theory and Practice, Chapman & Hall/CRC

❖ **Reference Books:**

1. William Stallings, Cryptography And Network Principles And Practice,Prentice Hall, Pearson Education Asia
2. AtulKahate, Cryptography & Network Security, The McGraw-Hill Companies
3. Behrouz A. Forouzan, Cryptography and Network Security, McGraw-Hill Companies

❖ **Reference Links/ e-content:**

1. http://people.csail.mit.edu/rivest/crypto-security.html
2. http://www.cryptix.org/
3. http://www.cryptocd.org/
4. http://www.cryptopp.com/

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER ENGINEERING

CS353: THEORY OF COMPUTATION

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	0	-	4	3
Marks	100	0	-	100	

Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1	Introduction	01
2	Finite Automata and Regular Languages	16
3	Non - Determinism Finite Automata	07
4	Context free Grammars	10
5	Pushdown Stack Memory Machines	06
6	Turing Machine	05

Total Hours (Theory): 45

Total Hours (Lab): 00

Total Hours: 45

Detailed Syllabus:

1.	**Introduction**	**01 Hours**	**5 %**
	Alphabets, Languages, State, length, Concatenation, Reverse, Null,Strings, Kleene closure, Positive Closure, Concept of Basic machine.		
2.	**Finite Automata and Regular Languages**	**16 Hours**	**26 %**
	Finite Automata and Regular Languages: Deterministic finite automata (DFA), Regular expressions, regular languages, Conversion of DFA to RE and RE to DFA Properties of regular sets applications, Automata with output - Moore machine, Mealy machine, Finite automata, memory requirement in a recognizer, definition, union, intersection and complement of regular languages.		
3	**Non - Deterministic Finite Automata**	**07 Hours**	**20 %**
	Non - Deterministic Finite Automata (NFA): Non - Deterministic Finite Automata, Conversion from NFA to FA, ^- Non Deterministic Finite Automata Conversion of NFA- ^ to NFA and equivalence of three Kleene'sTheorem, Minimization of Finite automata Regular And NonRegular Languages – pumping lemma.		
4.	**Context free Grammars**	**10 Hours**	**20 %**
	Context free grammar (CFG): Definition, Unions Concatenations And Kleen's of Context free language Regular grammar, Derivations and Languages, Relationship between derivation and derivation trees, Ambiguity Unambiguous CFG and Algebraic Expressions BacosNaur Form (BNF), Normal Form – CNF		
5.	**Pushdown Stack Memory Machines**	**06 Hours**	**14 %**
	Pushdown Automata, CFL And NCFL: Definition, deterministic PDA, Equivalence of CFG and PDA, Pumping lemma for CFL, Intersections and Complements of CFL, Non-CFL		

Course Outcome:

After completion of the course, Students will be able to:

CO1	Apply basic concepts of theory of computation in the computer field in order to solve computational problems.
CO2	Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation.
CO3	Analyze and design finite automata, pushdown automata and Turing machine for formal languages.
CO4	Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
CO5	Identify limitations of some computational models and possible solutions.
CO6	Design context free grammars for formal languages.

Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	-	-	-	-	-	-	-	-	3	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	2	-
CO3	3	2	2	-	-	-	-	-	-	-	-	-	3	-
CO4	2	2	2	-	-	-	-	-	-	-	-	-	1	-
CO5	2	1	1	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	-	-	-	-	-	-	-	-	-	-	-

Recommended Study Material:

❖ **Text Books:**

1. Introduction to Languages and Theory of Computation, John C. Martin, TMH

❖ **Reference Books:**

1. An introduction to automata theoryand formal languages, Adesh K. Pandey, S. K. Kataria & Sons
2. Introduction to computer theory, Deniel I. Cohen, John Wiley & Sons Inc
3. Computation: Finite and Infinite, Marvin L. Minsky, Prentice-Hall
4. "An introduction to Formal Languages and Automata", Peter Linz, 6th edition, Jones & Bartlett Learning
5. "Introduction to the Theory of Computation", Michael Sipser, 3rd edition, Cengage Learning.

❖ **Web Materials:**

1. http://en.wikipedia.org/wiki/Theory_of_computation

2. https://www.youtube.com/playlist?list=PLEbnTDJUr_IdM FmDFBJBz0zCsOF xfK
4. http://nptel.ac.in/courses/106103070/
5. http://nptel.ac.in/courses/106104028/
6. http://nptel.ac.in/courses/106106049/
7. https://www.youtube.com/watch?v=4GLC-s0PQLY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS362: COMPUTER NETWORKS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Outline of the Course:

Sr. No.	Title of the unit	Minimum Number of Hours
1	Introduction to Computer Networks	04
2	Data Link Layer	08
3	Medium Access Control Sub Layer	10
4	Network Layer	12
5	Transport Layer	08
6	Application Layer	03

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

Detailed Syllabus:

1	**Introduction to Computer Networks**	**04 Hours**	**09%**
1.1	Uses of computer network		
1.2	network hardware, network software		
1.3	OSI model, TCP/IP model, Comparison of OSI and TCP/IP model		
1.4	Example network		
2	**Data Link Layer**	**08 Hours**	**18%**
2.1	Design Issues		
2.2	framing, error control, flow control		
2.3	Error detection and correction		
2.4	Elementary data link protocols		
2.5	simplex, stop and wait, sliding window protocol, HDLC		
3	**Medium Access Control Sub Layer**	**10 Hours**	**22%**
3.1	The channel allocation problem, Multiple Access protocols: ALOHA, CSMA, Collision Free Protocols,		
3.2	Limited Contention Protocols, Wavelength Division Multiple Access Protocols		
3.3	Wireless LAN protocols; Ethernet: Traditional Ethernet, Switched Ethernet, Fast Ethernet, Gigabit Ethernet, IEEE 802.2: LLC, Datalink layer switching		
4	**Network Layer**	**12 Hours**	**27%**
4.1	Implementation of connection oriented and connection less service, Comparison of virtual circuit and datagram subnets, Routing algorithms		
4.2	Shortest path routing, Flooding, Distance vector routing, Link state routing, Hierarchical routing, Broadcast routing, Multicast routing, Routing for mobile host		
4.3	Routing in ad hoc network, Congestion control algorithms principles, Prevention policies		
4.4	Congestion control in virtual circuit subnets, Congestion control in datagram subnets, Load shedding,		
4.5	virtual circuit, Connectionless internetworking, Tunneling,Internetwork routing and fragmentation		
4.6	The network layer in the internet: The IP protocol, IPaddresses		
4.7	Internet control protocol, OSPF, BGP		

5	**Transport Layer**	**08 Hours**	**18%**
5.1	The transport service: Services provided to the upper layers		
5.2	Transport service primitives, Socket elements of transport protocols addressing		
5.3	Connection establishment, Connection release, Flow control		
5.4	Multiplexing, Crash recovery the transport protocol: UDP, TCP		
6	**Application Layer**	**03 Hours**	**06%**
6.1	DNS: The DNS name space, Resource records, Nameservers		
6.2	Electronic mail: Architecture and services		
6.3	World Wide Web: Architectural overview, HTTP.		

Course Outcome:

After completion of the course students will be able to

CO1	Analyze layered network architecture and passage of data over communication links
CO2	Analyze delay models in Data Networks using Queueing Systems for messaging and delay sensitive applications
CO3	Design and analyze routing algorithms for Internet and multi-hop autonomous networks
CO4	Analyze flow and rate control algorithms between a sender and receiver in wide area networks
CO5	Apply the network fundamentals to analyze performance.
CO6	Use key networking algorithms in simulation.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	-	-	-	-	-	-	-	2	-
CO2	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO3	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO4	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)Ifthere

is no correlation, put "-"

Recommended Study Material:

- **Text Books:**
  1. Computer Network, Andrew S. Tanenbaum, Prentice Hall PTR
- **Reference Books:**
  1. Introductionto Data Communication and Networking by Behrouz Forouzan, McGraw Hill
  2. Data and Computer Communications, William Stallings, Prentice Hall
- **Reference Books:**
  1. http://www.cisco.com
  2. http://compnetworking.about.com

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS364: Modern Artificial Intelligence

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours	3	2	-	5	**4**
Marks	100	50	-	150	

A. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction	04
2.	Problems, Problems Space and Search, Heuristic Search Techniques	10
3.	Knowledge Representation	04
4.	Neural Networks	10
5.	Reinforcement Learning	05
6.	NLP and Text Analytics:	07
7.	Advanced Topics	05

Total hours (Theory): 45
Total hours (Lab): 30
Total hours: 75

B. Detailed Syllabus:

1.	**Introduction**	**03 Hours**	**7%**
	What is AI, Applications of AI, characteristics, advantages and Disadvantages, Responsible AI		
2.	**Problems, Problems Space and Search, Heuristic Search Techniques**	**05 Hours**	**20%**
	Defining The Problems as a State Space Search, Production Systems, Problem Characteristics, Production System Characteristics, Issues In The Design Of Search Programs, Heuristic Search Techniques: Hill Climbing, A*, AO*, Simulated Annealing, Branch and Bound, Nearest Neighbor, Blind Search Techniques: DFS, BFS, Best First Search, Control Strategies.		
3.	**Knowledge Representation**	**04 Hours**	**10%**
	Propositional Logic, Predicate Logic, Monotonic and non-Monotonic, Knowledge Representations, Approaches to Knowledge Representation, Procedural Versus Declarative Knowledge.		
4.	**Neural Networks**	**10 Hours**	**20%**
	Neurons and biological motivation, Linear threshold units, Perceptron's Algorithm, Activation Functions, Neural Network design, Multilayer networks, Feed Forward and Backpropagation Learning Algorithm, Types of Neural Network Architecture, Bias-Variance trade-off. Regularization and model/feature selection, Optimizers (Gradient Descent, Adagrad, RMSProp, Adam), Learning Rate Introduction to Deep Neural Network, Convolution Neural Network (CNN), Recurrent Neural Network (RNN), Generative Adversarial Networks (GAN)		
5.	**Reinforcement Learning**	**12 Hours**	**15%**
	What is RL and their types, Markov Decision Processes (MDPs), Policies and value functions Learning optimal policies and value functions, Q-learning, Introduction to Q-learning with value iteration, Implementing an epsilon greedy strategy		

	Deep Q-networks (DQNs), Introduction to DQNs, Create and train DQN to play game, watch trained DQN play game, Policy gradients.		
6.	**NLP and Text Analytics:**	**06 Hours**	**15%**
	Introduction, Syntactic Processing, Semantic Analysis, Semantic Analysis, Discourse And Pragmatic Processing, Text Analytics, Text pre-processing, Bag of Words, Word Cloud, Machine Translation, sentiment analysis		
7.	**Advanced Topics**	**06 Hours**	**13%**
	Expert System and Optimization Techniques, Conversational AI, Explainable AI, Data Centric AI, Recent Advancement in AI and Case study: Agriculture, Healthcare, Transportation, Education, Robotics, E-commerce, Finance, Data Security.		

C. Instructional Methods and Pedagogy:

- At the start, of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Faculty would use coached problem-solving method as it is the class format in which faculty provide a structured, guided context for students working collaboratively to solve problems.
- Assignments based on topic content will be given to the students at the end of each unit/topic.
- Surprise tests/Quizzes will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

D. Student Learning Outcomes / Objectives:

Upon completion of this course, students will be able to do the following:

- This subject will help students to solve difficult and complex problem of computer science using AI techniques.
- AI knowledge will help student to select any R&D field related to application of AI in PG courses.
- Basic knowledge of AI will let students to understand soft computing and machine learning.
- AI techniques will be utilized to develop software solution as per need of today's IT edge which requires high automation and less human intervention.

E. Recommended Study Material:

❖ **Text Books:**

1. "Artificial Intelligence" -By Elaine Rich and Kevin Knight (2nd Edition) Tata Mcgraw-Hill.
2. Stuart J. Russell and Peter Norvig, Artificial Intelligence 3e: A Modern Approach, 3rd Edition. Person
3. Deep Learning, by Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press.
4. Computer Vision: Algorithms and Applications, 2nd ed., by Richard Szeliski, The University of Washington

❖ **Reference Books:**

1. "Artificial Intelligence and Expert System, Development"-By D.W.Rolston, Mcgraw-Hill International Edition.
2. "Artificial Intelligence And Expert Systems" By D.W.Patterson

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS378: VIRTUAL & AUGMENTED REALITY (PE-II)

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- Introduction to Computer Programming

Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1.	Introduction of Virtual Reality (VR) and Augmented Reality(AR)	4
2.	Multiple Modals of Input and Output Interface in Augmented Reality	4
3.	Geometry of Virtual Worlds	4
4.	Light and Optics	4
5.	Visual Physiology and Perception	5
6.	Tracking System	4
7.	Interactive Techniques in Augmented Reality	4
8.	Introduction of Immersive VR Hardware	4
9.	Development Tools and Frameworks in Virtual Reality	4
10.	Application of AR/VR	4
11.	Discussion over research paper in Area of VR/AR	4

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

Detailed Syllabus:

1	**Introduction of Augmented reality**	**04 hours**	**09%**
1.1	Fundamental Concept and Components of AR/VR		
1.2	Primary Features and Present Development on AR/VR		
1.3	Birds-eye view (general), Birds-eye view (hardware)		
1.4	Birds-eye view (software), Birds-eye view (sensation and perception)		
2	**Multiple Modals of Input and Output Interface in Augmented Reality**	**04 hours**	**09%**
2.1	Input -- Tracker, Sensor, Digital Glove, Movement Capture		
2.2	Input --Video-based Input, 3D Menus & 3DScanner		
2.3	Output -- Visual / Auditory / Haptic Devices		
3	**Geometry of Virtual Worlds**	**04 hours**	**09%**
3.1	Geometric modeling, Transforming models		
3.2	Matrix algebra and 2D rotations		
3.3	3D rotations and yaw, pitch, and roll		
3.4	Axis-angle representations, Quaternions		
3.5	Converting and multiplying rotations, Homogeneous transforms		
3.6	The chain of viewing transforms, Eye transforms		
4	**Light and Optics**	**04 hours**	**09%**
4.1	Three interpretations of light		
4.2	Refraction, Simple lenses		
4.3	Diopters, Imaging properties of lenses		
4.4	Lens aberrations		
4.5	Optical system of eyes		
5	**Visual Physiology and Perception**	**5 hours**	**12%**
5.1	Photoreceptors, Sufficient resolution for VR		
5.2	Light intensity, Eye movements		
5.3	Neuroscience of vision		
5.4	Depth perception, Motion perception		
5.5	Frame rates and displays		
6	Tracking System	04 Hour	**08%**
6.1	Orientation Tracking, Tilt and Yaw drift Correction		

6.2	Tracking with Camera, Filtering		
7	**Interactive Techniques in Virtual Reality**	**04 hours**	**09%**
7.1	Body Track, Hand Gesture		
7.2	3D Manus		
7.3	Object Grasp		
8	Introduction of Immersive VR Hardware	**04 hours**	**09%**
8.1	VR Box, HMD		
8.2	CAVE		
8.3	**Data Gloves, Controller.**		
9	**Development Tools and Frameworks in Virtual Reality**	**04 hours**	**09%**
9.1	Frameworks of Software Development Tools in VR		
9.2	Unreal Engine, Unity3D		
9.3	Blender, A-Frame		
10	**Application of AR/VR in D**	**04 hours**	**09%**
10.1	VR Technology in Film & TV Production, In health Care		
10.2	VR Technology in Physical Exercises, Games, Education		
10.3	Demonstration of VR applications		
11	**Discussion over research paper in Area VR/AR**	**04 hours**	**08%**

Course Outcome:

After completion of the course students will be able to

CO1	Describe how AR/VR systems work and list the applications of AR/VR.
CO2	Understand the design and implementation of the hardware that enables AR/VR systems to be built.
CO3	Understand the system of human vision and its implication on perception and rendering
CO4	Explain the concepts of motion and tracking in VR system
CO5	Describe the importance of interaction and audio in VR systems.
CO6	Develop the creativity and designing skill in 3D development.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	-	-	3		3	-	-	-	-	-	3	-
CO2	-	-	-	-	-	3	-	-	-	-	2	-	3	-
CO3		3	-	2	-	-	-	1	-	3	-	-		-
CO4	3		1	-	-	-	-		-	-	-	-	2	-
CO5	-	-	-	-	-	-	-	-	2		3	-	1	-
CO6	-	-	-	-	-	3	-	-	-	-	-	2	-	-

Recommended Study Material:

❖ **Text Books:**

1. Virtual Reality (The MIT Press Essential Knowledge series) by Samuel Greengard.
2. VIRTUAL REALITY By Steven M. LaValle Cambridge University Press.

❖ **Reference Books:**

1. Virtual Reality with VRTK4: Create Immersive VR Experiences Leveraging Unity3D and Virtual Reality Toolkit By Rakesh Baruah.
2. Virtual Reality(The MIT Press Essential Knowledge series) by Samuel Greengard
3. George Mather, Foundations of Sensation and Perception: Psychology Press; 2 edition, 2009.
4. Peter Shirley, Michael Ashikhmin, and Steve Marschner, Fundamentals of Computer Graphics, A K Peters/CRC Press; 3 edition, 2009.
5. Burdea G. C., Coiffet P., "Virtual Reality Technology", J. Wiley & Sons, Second Ed., 2003.
6. Watt A., Policarpo F., "3D Games. Real-time Rendering and Software Technology". Addison-Wesley, 2001.

❖ **Web Materials:**

1. http://nptel.ac.in
2. http://msl.cs.uiuc.edu/vr/
3. www.coursera.com
4. http://cgis.utcluj.ro/teaching/

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS379: WIRELESS COMMUNICATION AND MOBILE COMPUTING (PE-II)

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Pre-requisite courses:

- Computer Networks
- Modern Networks

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Wireless Communication Fundamentals	03
2.	Telecommunication Systems	12
3.	Wireless LAN	18
4.	Mobile Network Layer	12
5.	Mobile Transport Layer	12
6.	Advanced Mobile Technologies	03

Total hours (Theory): 45 Hrs.

Total hours (Lab): 30 Hrs.

Total hours: 75 Hrs.

Detailed Syllabus:

1.	**Wireless Communication Fundamentals**	**03 hours**	**05%**
1.1	Applications		
1.2	A short history of wireless communication.		
1.3	A market for mobile communications		
1.4	Some open research topics		
1.5	A simplified reference model		
2.	**Telecommunication Systems**	**12 hours**	**21%**
2.1	GSM: Mobile services, System architecture, Radiointerface, Protocols, Localization and Calling, Handover, Security, New data services		
2.2	DECT: System architecture, Protocol architecture		
2.3	TETRA, UMTS and IMT–2000		
3.	**Wireless LAN**	**18 Hours**	**28 %**
3.1	Infrared vs radio transmission		
3.2	Infrastructure and ad-hoc network		
3.3	IEEE 802.11: System architecture, Protocol architecture, Physical layer, medium access controllayer, MAC management, 802.11a, 802.11b, Newer developments		
3.4	HiperLAN: HiperLAN 1, HiperLAN2		
3.5	Bluetooth: Architecture, Radio layer, Baseband layer, Link Manager Protocol, L2CAP, Security, SDP,Profiles, IEEE 802.15		
4.	**Mobile Network Layer**	**12hours**	**21%**
4.1	Mobile IP		
4.2	Dynamic Host Configuration Protocol		
4.3	Mobile ad–hoc networks: Routing, Destination sequencedistance vector, Dynamic source routing		
5.	**Mobile Transport Layer**	**12 hours**	**21%**
5.1	Traditional TCP: Congestion control, Slow start, Fast retransmit/fast recovery,Implications of mobility		
5.2	Classical TCP improvements: Indirect TCP, Snooping TCP, Mobile TCP, Fastretransmit/fast recovery, Transmission/time- out freezing, Selective retransmission, Transaction-oriented TCP		
5.3	TCP over 2.5/3G wireless networks		
6.	**Advanced Mobile Technologies**	**03 hours**	**04%**

6.1	Introduction of 4G, 5G and other advanced mobile technologies	
6.2	The architecture of future networks	

Course Outcome:

After completion of the course, Students will be able to:

CO1	Demonstrate the principles and theories of mobile computing technology
CO2	Characterize the infrastructure and technology of mobile computing technologies
CO3	List applications in various domains offered to the public, employees and businesses through mobile computing technology
CO4	Describes the current technologies to build potential future of mobile computing technologies and applications
CO5	Identify the potential future of mobile computing technologies and applications

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO1	3	1	-	-	-	-	-	-	-	-	-	-
CO2	1	2	1	-	-	-	-	-	-	-	-	-
CO3	1	2	2	-	2	-	-	-	-	-	1	-
CO4	-	3	1	1	-	-	1	-	-	-	-	-
CO5	-	3	1	1	-	-	1	-	-	-	-	-

Recommended Study Material:

- **Text Books:**

1. Jochen Schiller, "Mobile Communications", PHI/Pearson Education, Second Edition, 2003
2. "Mobile Computing: Technology, Applications and Service Creation" by Asoke K Talukder and Roopa R Yavagal, TMH,ISBM: 0-07-058807-4

- **Reference Books:**

1. William Stallings, "Wireless Communications and Networks", PHI/Pearson Education, 2002.
2. Kaveh Pahlavan, Prasanth Krishnamoorthy, "Principles of Wireless Networks",
3. PHI/Pearson Education, 2003.
4. Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, "Principles of

5. Mobile Computing”, Springer, New York, 2003.
6. Hazysztof Wesolowshi, “Mobile Communication Systems”, John Wiley and Sons Ltd, 2002
7. Research papers from IEEE, Springer etc.

❖ **Reference Links/ e-content:**

1. www.ietf.org – For drafts
2. www.ieee.org – For standards and technical research paper

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS383: CYBER SECURITY AND CYBER LAWS

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

A. Objective of the Course:

The main objective to give the course is

- To be able to identify vulnerabilities in system, software or web application
- To be able to identify security risks and take preventive steps
- To be able investigate cybercrime and collect evidences by understanding cyber laws.
- Able to use knowledge of forensic tools and software

B. Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1	Introduction to Cyber Security	06
2	Cyber Security Vulnerabilities and Cyber Security Safeguards	11
3	Securing Web Application, Services and Servers	06
4	Intrusion Detection and Prevention	11
5	Introduction to Cyber Crime and law	06
6	Introduction to Cyber Crime Investigation	05

Total Hours (Theory): 45

Total Hours (Lab): 30

Total Hours: 75

Detailed Syllabus:

1	**Introduction to Cyber Security**	**06 hours**	**12%**
1.1	Overview of Cyber Security		
1.2	Internet Governance – Challenges and Constraints		
1.3	Cyber Threats: - Cyber Warfare-Cyber Crime-Cyber terrorism-Cyber Espionage		
1.4	Need for a Comprehensive Cyber Security Policy, Need for a Nodal Authority, Need for an International convention on Cyberspace.		
2	**Cyber Security Vulnerabilities and Cyber Security Safeguards**	**11 hours**	**25%**
2.1	Cyber Security Vulnerabilities-Overview		
2.2	Vulnerabilities in software, System administration, Complex Network Architectures, Open Access to Organizational Data, Weak Authentication, Unprotected Broadband communications, Poor Cyber Security Awareness.		
2.3	Cyber Security Safeguards- Overview		
2.4	Access control, Audit, Authentication, Biometrics, Cryptography, Deception, Denial of Service Filters		
2.5	Ethical Hacking, Firewalls, Intrusion Detection Systems, Response, Scanning, Security policy, Threat Management.		
3	**Securing Web Application, Services and Servers**	**06 hours**	**13%**
3.1	Introduction, Basic security for HTTP Applications and Services, Basic Security for SOAP Services		
3.2	Identity Management and Web Services, Authorization Patterns		
3.3	OWASP Top 10, Security Considerations, Challenges		
4	**Intrusion Detection and Prevention**		
4.1	Introduction of Intrusion Detection and Prevention, Physical Theft, Abuse of Privileges, Unauthorized Access by Outsider, Malware infection		
4.2	Intrusion detection and Prevention Techniques, Anti-Malware software		
4.3	Network based Intrusion detection Systems, Network based Intrusion Prevention Systems		
4.4	Host based Intrusion prevention Systems, Security Information Management		
4.5	Network Session Analysis, System Integrity Validation		
5	**Introduction to Cyber Crime and law**	**06 hours**	**13%**

5.1	Cyber Crimes, Types of Cybercrime, Hacking, Attack vectors		
5.2	1Cyber Crimes, Types of Cybercrime, Hacking, Attack vectors		
5.3	Cyberspace and Criminal Behaviour, Clarification of Terms, Traditional Problems Associated with Computer Crime		
5.4	Introduction to Incident Response, Digital Forensics, Realms of the Cyber world, Recognizing and Defining Computer Crime		
5.5	Contemporary Crimes, Contaminants and Destruction of Data 1.1 Indian IT ACT 2000		
6	**Introduction to Cyber Crime Investigation**	**05 hours**	**12%**
6.1	Key loggers and Spyware, Virus and Warms, Trojan and backdoors		
6.2	DOS and DDOS attack		
6.3	Steganography, SQL injection, Buffer Overflow		

C. Instructional Methods and Pedagogy:

- At the start, of course, the course delivery pattern, prerequisite of the subject will bediscussed.
- Faculty would use coached problem-solving method as it is the class format in which faculty provide a structured, guided context for students working collaboratively to solveproblems.
- Assignments based on topic content will be given to the students at the end of eachunit/topic.
- Surprise tests/Quizzes will be conducted.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

D. Student Learning Outcomes / Objectives:

Upon completion of this course, students will be able to do the following:

- After learning the course, students should able to learn about cyber-attack, type of cybercrimes
- Learn cyber security vulnerabilities and security safeguard.
- Learn hand-on on how to prevent the web using different web application tools.
- Learn about how to protect the network using different network tools and Intrusiondetection and prevention techniques,
- Learn basic of Cryptography and network security
- Understand about cyber laws
- Learn about cyber Forensics and investigation of information hiding.
- Learn about how to protect them self and ultimately society from cyber-attacks.

E. Recommended Study Material:

❖ Reference Text Books:

1. Digital Privacy and Security Using Windows: A Practical Guide ByNihad Hassan, RamiHijazi, Apress
2. Cyber Security Understanding Cyber Crimes, Computer Forensics and Legal Perspectivesby Nina Godbole and SunitBelpure, Publication Wiley
3. Cybersecurity: The Beginner’s Guide Dr.Erdal Ozkaya

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS357: SOFTWARE GROUP PROJECT -IV

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	100	100	

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most ***three***) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
  a. Project Synopsis
  b. Software Requirement Specification
  c. SPMP
  d. Final Project Report
  e. Project Setup file with Source code
  f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total Hours (Theory): 00

Total Hours (Lab): 60

Total Hours: 60

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Identify and define the computing requirements of a problem to propose its appropriate solution.
CO2	Correlate knowledge of different subjects and apply it to implement solution of the problem.
CO3	Apply engineering and management principles to achieve project goal.
CO4	Prepare technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO5	Able to communicate effectively with a range of audiences.
CO6	Recognition of the need for and an ability to engage in continuing professional development.
CO7	Able to work and coordinate with different kind of people in the team

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	2	-	-	-	3	-
CO2	3	-	1	1	3	1	1	-	-	-	-	2	1	1
CO3	-	-	3	-	3	1	1	-	2	-	3	2	3	-
CO4	-	-	-	-	3	-	2	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	2	-	2

Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics

3. Various software tools and programming languages compiler related to topic

- **Web Materials:**
  1. www.ieeexplore.ieee.org
  2. www.sciencedirect.com
  3. www.elsevier.com
  4. http://spie.org/x576.xml

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS132.02 A: CONTRIBUTOR PERSONALITY DEVELOPMENT (HSSELECTIVE-III)

Credits and Schemes:

<table>
<tr><th rowspan="3">Sem.</th><th rowspan="3">Course Code</th><th rowspan="3">Course Name</th><th rowspan="3">Credits</th><th rowspan="3">Teaching Scheme Contact Hours/Week</th><th colspan="4">Evaluation Scheme</th><th rowspan="3">Total</th></tr>
<tr><th colspan="2">Theory</th><th colspan="2">Practical</th></tr>
<tr><th>Internal</th><th>External</th><th>Internal</th><th>External</th></tr>
<tr><td>VI</td><td>HS132.02 A</td><td>CONTRIBUTOR PERSONALITY DEVELOPMENT</td><td>02</td><td>02</td><td>--</td><td>--</td><td>30</td><td>70</td><td>100</td></tr>
</table>

Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Concept of Personality:** • Meaning of Personality • Types of Personality • Factors contributing to Personality • Personality Traits	06
2	**Soft Skills and Personality Development:** • Critical, Creative and Positive Thinking • Leadership, Assertiveness and Negotiation Skills • Self-Management • People's Skills • Building Relationship Skills • Being a Team Player	08
3	**Developing Contributor Personality – Part I** • Concept of Contributor • Characteristics of a Contributor • The Contributor's Vision of Success & Career • The Scope of Contribution • Embarking on the Journey to Contributor ship	06

4	**Developing Contributor Personality – Part II** • Focus on Values • Engage Deeply • Think in Enlightened Self-Interest • Practice Imaginative Sympathy • Demonstrate Trust Behavior • Developing a sense of duty and morality	06
5	**Contemporary Issues in CPD** • Contemporary Practices & Trends in Contributor Personality Development • Case Study & Presentations	04
	Total	**30**

Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, critical reading, group work, independent and collaborative research, presentations etc.

Evaluation

The students will be evaluated continuously in the form of internal as well as external examinations. The practical evaluation is schemed as 30 marks for internal evaluation and 70 marks for external evaluation in the form of University examination.

Internal Evaluation

The students' performance in the course will be on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
3	Assignment / Project Work / Term Work / Quiz	2	25	25
4	Attendance and Class Participation			05
			Total	30

External Evaluation

The University Practical examination will be of 70 marks and will test the contributory personality aspects and their applications by carrying out practical assessment. The examination will avoid, as far as possible, grammatical errors and will focus on applications.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Practical Exam / Viva	01	70	70
			Total	70

Course Outcome (COs):

After completion of the course, the student would:

CO1	Develop conceptual understanding of the term Personality and understand one's own personality Traits
CO2	Develop soft skills for holistic personality development for career and in personal life
CO3	Develop assertiveness, self-management and negotiation skills to navigate the personal and professional environment successfully
CO4	Develop conceptual clarity of the term 'Contributor Personality' and develop understanding of the traits of a contributor
CO5	Develop the qualities of a contributor such as trust, strong work ethic, responsibility and accountability at workplace
CO6	Develop relevant skills required to be a contributor at workplace and achieve success in life

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO3	-	-	-	-	-	-	-	-	3	-	-	-	-	-
CO4	-	-	-	-	-	-	-	-	-	-	-	2	-	-
CO5	-	-	-	-	-	-	-	-	-	2	-	-	-	-
CO6	-	-	-	-	-	2	-	-	3	-	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

- 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books / Reading

- Contributor Personality Program Workbook (Volume 1,2),
- Contributor Personality Program ActivGuide, Illumine Knowledge Pvt. Ltd.

B. Tech. (Computer Science & Engineering) Programme

SYLLABI

(Semester –7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OFCOMPUTER SCIENCE & ENGINEERING

CS449: INTERNET OF THINGS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	4
Marks	100	50	-	150	

Pre-requisite courses:

- **Computer Network**
- **Wireless Communication**
- **Embedded system**

Outline of the course:

Sr. No.	Title of the unit	Minimum number of hours
1	Introduction and Evolution of IoT	04
2	Organization and primary components of IoT systems	08
3	A reference IoT architecture	10
4	Design issues for the IoT edge	12
5	Security, trust, and privacy issues in IoT	08
6	IoT case studies	03

Total hours (Theory): 45
Total hours (Lab): 30
Total hours: 75

Detailed Syllabus:

1	**Introduction and evolution of IoT**	**04 Hours**	**09%**
1.1	Internet of Things Definition Evolution, Origin, Definition,Characteristics , applications, need and scope of IoT, functional stack, Cisco IoT Architecture, Processors and Operating Systems for resource constraineddevices , Sensors and actuators, smart objects, IoT vs M2M, IoT vs WoT, IoE.		
2	**Organisation and primary components of IoT systems**	**08 Hours**	**18%**
2.1	Structure of IoT systems		
2.2	IoT backend modules		
2.3	IoT gateways, IoT Cloud platforms : AWS IoT Platform, Azure IoT Platform, IBM Bluemix Platform, Sensor-Cloud		
2.4	Edge Computing, Fog Computing		
3	**A reference IoT architecture**	**10 Hours**	**22%**
3.1	Design principles and design requirements for the reference architecture		
3.2	Real-world constraints		
4	**Design issues for the IoT edge**	**12 Hours**	**27%**
4.1	Sensors and actuators for IoT systems		

4.2	Interoperability and reliability issues		
4.3	Communication protocols and protocol stacks for the edge devices (HTTP, CoAP, MQTT, AMQP, XMPP)		
4.4	Hardware security for edge devices		
5	**Security, trust, and privacy issues in IoT**	**08 Hours**	**18%**
5.1	Identity management of IoT edge devices		
6	**IoT case studies**		**03 Hours**
6.1	Smart grid		
6.2	Home automation		
6.3	Industrial IoT		

Instructional Method and Pedagogy:

- At the start of course, the course delivery pattern, prerequisite of the subject will be discussed.
- Lectures will be conducted with the aid of multi-media projector, black board, OHP etc.
- The course includes a laboratory, where students have an opportunity to build an appreciation for the concepts being taught in lectures.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Analyze and utilization of IoT for latest trend in IT sector.
CO2	Provide an understanding of the technologies and the standards relating to the Internet of Things.
CO3	Integration of Existing technology for development of IoT Applications

CO4	Student will be able to make program which works on Sensors
CO5	Addressing security, privacy and standardisation issues in implementation of IoT.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	-	2	-	-	-	-	-	-	-	1	-	-	-
CO2	-	3	2	-	3	-	-	3	-	3	-	-	2	-
CO3	-	2	-	3	2	1	-	-	1	-	-	-	-	-
CO4	-		-	2	2	-	3	-	-	2	-	-	-	2
CO5	1	-	-	-	3	2	-	-	3	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put “-”

Recommended Study Material:

Text book:

1. Internet of Things: Architectures, Protocols and Standards 1st Edition , Simone Cirani, Gianluigi Ferrari, Marco Picone, and Luca Veltri.
2. Internet of Things: principles and paradigms, Buyya, Rajkumar and Amir Vahid Dasterdji (eds.), Morgan Kaufmann, 2016.
3. From Machine-to-Machine to the Internet of Things: introduction to a new age of intelligence, Holler, Jan et al, Academic Press, 2014.

Reference book:

1. The Internet of Things Enabling Technologies, Platforms, and Use Cases, Pethuru Raj Anupama **A.** Raman,2017
2. Building Internet of Things with the Arduino, Doukas, Charalampos, Create Space Independent Publishing Platform, 2012.
3. Francis daCosta, “Rethinking the Internet of Things: A Scalable Approach to Connecting Everything”, 1st Edition, Apress Publications, 2013.

Web material:

1. http://web.mit.edu/professional/digital-programs/courses/IoT/phone/index.html
2. https://swayam.gov.in/nd1_noc19_cs65/preview

3. https://www.edureka.co/blog/iot-tutorial/
4. http://www.steves-internet-guide.com/internet-of-things/

Software:

1. Contiki OS
2. Node-Red
3. Proteus
4. Thinker Cad

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS442: DATA SCIENCE & ANALYTICS

Credits and Hours:

Teaching Scheme	**Theory**	**Practical**	**Total**	**Credit**
Hours/week	3	4	7	**5**
Marks	100	100	200	

Pre-requisite courses:

- Data Structures & Algorithm Design
- Database Management System
- Design & Analysis of Algorithms
- Computer Programming
- Engineering Mathematics

A. Outline of the Course:

Sr. No.	**Title of the unit**	**Minimum number of hours**
1.	INTRODUCTION TO DATA SCIENCE	04
2.	STATISTICAL INFERENCE	05
3.	DATA PRE-PROCESSING AND DATA VISUALIZATION	05
4.	INTRODUCTION TO MAP-REDUCE AND HADOOP ARCHITECTURE	05
5.	HDFS, HIVE AND HIVEQL, HBASE	10
6.	APACHE SPARK	06
7.	NoSQL	03
8.	DATA BASE FOR THE MODERN WEB	07
	Total hours (Theory):	45
	Total hours (Lab):	60
	Total hours:	105

B. Detailed Syllabus:

1.	**INTRODUCTION TO DATA SCIENCE**	**04 Hours**	**10%**
	Introduction of data science and data analytics, defining data science by its key components, Big Data and its importance, Four Vs, Drivers for Big data, Big data applications, Exploring Data Science in Business, Applications in real-world		
2.	**STATISTICAL INFERENCE**	**05 Hours**	**08%**
	Event Space, Random Variables and Probability Distributions		
3.	**DATA PRE-PROCESSING AND DATA VISUALIZATION**	**05 Hours**	**15%**
	Dataset, Types of Dataset, Importance of Pre-processing the Data Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation Data visualization techniques		
4.	**INTRODUCTION TO MAP-REDUCE AND HADOOP ARCHITECTURE**	**08 Hours**	**25%**
	Big Data – Apache Hadoop & Hadoop EcoSystem, Moving Data in and out of Hadoop – Understanding inputs and outputs, MapReduce, Data Serialization.		
5.	**HDFS, HIVE AND HIVEQL, HBASE**	**10 Hours**	**20 %**
	HDFS-Overview, Installation and Shell, Java API; Hive Architecture and Installation, Comparison with Traditional Database, HiveQL Querying Data, Sorting And Aggregating, Map Reduce Scripts, Joins & Sub queries, HBase concepts, Advanced Usage, Schema Design, Advance Indexing, PIG, Zookeeper , how it helps in monitoring a cluster, HBase uses Zookeeper and how to Build Applications with Zookeeper.		
6.	**Apache SPARK**	**06 Hours**	**15%**
	Introduction to Data Analysis with Spark, Downloading Spark and GettingStarted, Programming with RDDs, Machine Learning with MLlib.		

7.	**NoSQL**	**03 Hours**	**08 %**
	What is it? Where It is Used Types of NoSQL databases, Why NoSQL? Advantages of NoSQL, Use of NoSQL in Industry, SQL vs NoSQL, NewSQL		
8.	**Data Base for the Modern Web**	**07 Hours**	**12 %**
	Introduction to MongoDB key features, Core Server tools MongoDB through the JavaScript's Shell, Creating and Querying through Indexes, Document-Oriented, principles of schema design Constructing queries on Databases, collections and Documents MongoDB Query Language.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Use an ethically responsible approach to evaluate and interpret data
CO2	Demonstrate expertise in statistical data processing
CO3	Use of various algorithms as well as mathematical and statistical models and optimization concepts to formulate and the use analyse data appropriately
CO4	Develop the ability to build and evaluate data-based models.
CO5	To learn difference between conventional SQL query language and NoSQL andMongoDBbasic concepts
CO6	Utilizing data science principles and approaches to solve real-life situational problems and effectively communicate them.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12
CO1	3	2	1	2	-	-	-	-	-	-	-	-
CO2	1	3	1	-	-	-	-	-	-	-	-	-
CO3	-	2	3	2	-	-	-	-	-	-	-	-
CO4	-	-	3		-	-	-	-	-	-	-	-
CO5	-	2	2	3	-	-	-	-	-	-	-	-
CO6	-	-	-	-	-	-	-	1	-	-	-	3

Recommended Study Material:

- **Text book:**
  1. Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data, byEMC Education Services, Wiley, 2015.
  2. Professional Hadoop Solutions ByBoris lublinsky, Kevin t. Smith, Alexey Yakubovich, Wiley, ISBN: 9788126551071, 2015.
  3. Understanding Big data ByChris Eaton,Dirkderooset al. , McGraw Hill,2012.
  4. BIG Data and Analytics ,Sima Acharya, Subhashini Chhellappan, Willey
  5. MongoDB in Action, Kyle Banker,PiterBakkum , Shaun Verch, Dream tech Press
  6. HADOOP: The definitive Guide By Tom White, 4th Edition O Reilly 2012.
  7. Big Data Analyticswith R and Haoop ByVigneshPrajapati, Packet Publishing 2013.
  8. Learning Spark: Lightning-Fast Big Data Analysis Paperback by Holden Karau, Apress
- **Reference book:**
  1. Big Data Analytics with Spark ByGuller, Mohammed,Apress
  2. Analytics in a Big Data World: The Essential Guide to Data Science and Its Applications By Bart Baesens, Wiley Publication
  3. Hadoop in Practice by Alex Holmes, Manning Publication
- **Web material:**
  1. http://www.bigdatauniversity.com/
  2. https://sparkhub.databricks.com/resources/

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS450: Design of Language Processors

Credit and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	
Marks	100	50	-	150	**4**

Pre-requisite courses:

- Digital Electronics
- Operating System
- Theory of Computation

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of Language Processors & Lexical Analysis	08
2.	Syntax Analysis	09
3.	Parsing Methods	10
4.	Syntax-Directed Translation & Intermediate Code Generation	08
5.	Runtime Environment & Code Generation	10
	Total hours (Theory):	45
	Total hours (Lab):	30
	Total hours:	70

Detailed Syllabus:

1.	**Overview of Language Processors & Lexical Analysis**	**08 Hours**	**15%**
	• Language Processors • The Structure if a Compiler • Application of Compiler Technology **Lexical Analysis:** • The Role of Lexical Analyzer • Specification of Tokens • Recognition of Tokens • Lexical Analyzer Generator LEX		
2.	**Syntax Analysis**	**09 Hours**	**15%**
	• Role of the Parser • Representative Grammar • Syntax Error Handling • Error-recovery Strategies		
3.	**Parsing Methods**	**10 Hours**	**30%**
	• Top Down Parsing: Recursive-Descent Parsing, FIRST and FOLLOW, LL(1)grammar • Non-recursive Predictive Parsing • Construction of Non-recursive Predictive Parsing Table • Error Recovery in Predictive Parsing • Bottom-up Parsing: Shift-Reduce Parsing, Conflicts during Shift-ReduceParsing • Introductionto LR Parsing, L-R Parsing Algorithm, Viable Prefixes • Simple LR Parser (SLR), Construction of Simple LR Parsing Table • Canonical LR(1), Construction of LR(1) Parsing Table • Look Ahead LR (LALR), Construction of LALR Parsing Table • Parser Generator – Yacc		

4.	**Syntax-Directed Translation & Intermediate Code Generation**	**08 Hours**	**25%**
	• Syntax-Directed Definitions • Dependency Graphs • S-attributed Definitions • L-attributed Definitions • Application of Syntax Directed Translation • Syntax Directed Translation Schemes **Intermediate Code Generation:** • Variants of Syntax Trees • Three Address Code • Control Flow		
5.	**Runtime Environment & Code Generation**	**10 Hours**	**15%**
	• Storage Organization • Activation Trees • Activation Records • Calling Sequence • Heap Management • Introduction to Garbage Collection **Code Generation** • Issues in Code Generator • The Target Language • Basic Blocks and Flow Graphs • Optimization of Basic Blocks • A simple Code Generator • Peephole Optimization		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand fundamentals of compiler and identify the relationships among different phases of the compiler and use the knowledge of the Lex tool
CO2	Describe Role of Parser and the various error recovery strategies
CO3	Develop the parsers and experiment with the knowledge of different parsers design.
CO4	Design syntax directed translation schemes for a given context free grammar
CO5	Develop semantic analysis scheme to generate intermediate code
CO6	Summarize various optimization techniques used for dataflow analysis and generate machine code from the source code

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	2	-	-	-	-	-	-	-	-	-
CO2	-	2	-	1	-	-	-	-	-	-	-	-	-	-
CO3	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO4	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO5	-	1	-	2	-	-	-	-	-	-	-	-	-	-
CO6	-	-	2	1	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- **Text book:**
  1. Alfred Aho, Ravi Sethi, Jeffrey D Ullman, "Compilers Principles, Techniques and- Tools", Pearson Education Asia.
  2. M. Dhamdhere, "System Programming and Operating Systems", Tata McGraw-Hill.
  3. Steven S. Muchnick. Advanced Compiler Design and Implementation
- **Reference book:**
  1. Allen I. Holub "Compiler Design in C", Prentice Hall of India.
  2. C. N. Fischer and R. J. LeBlanc, "Crafting a compiler with C", Benjamin

Cummings.

3. J.P. Bennet, “Introduction to Compiler Techniques”, Second Edition, Tata McGraw-Hill
4. HenkAlblas and Albert Nymeyer, “Practice and Principles of Compiler Building with C”, PHI.
5. Kenneth C. Louden, “Compiler Construction: Principles and Practice”, Thompson Learning.
6. Compiler Construction by Kenneth. C. Louden, Vikas Pub

❖ **Web material:**

1. http://compilers.iecc.com/crenshaw
2. http://www.compilerconnection.com
3. http://dinosaur.compilertools.net
4. http://pltplp.net/lex-yacc

❖ **Software:**

1. LEX
2. YACC

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS451: ADVANCED COMPUTING

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	3	2	5	4
Marks	100	50	150	

Pre-requisite courses:

- Operating System
- Networking

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Introduction to Computing Technology	08
2.	Cloud Enabling Technologies	08
3.	Cloud Architectures	08
4.	Edge Computing & its Applications	06
5.	Fog Computing & its Applications	06
6.	Container Technology & Tools	06
7.	Market Place of Advanced Computing Platforms	03
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1	**Introduction to Computing Technology** Overview of Cluster Computing, Grid Computing Systems, Cloud Computing, Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, Cloud Deployment Models, Desired Features of a Cloud, Benefits and Disadvantages of Cloud Computing, Challenges and Risks in Cloud Computing.	**08 Hours**	**15%**
2	**Cloud Enabling Technologies** Data Center Technology, Virtualization Technology, Implementation Levels of Virtualization, Virtualization Structures/Tools and Mechanisms, Managing Virtualization Environment, Types of Hypervisors, Virtualization of CPU, Memory, and I/O Devices, Virtual Clusters and Resource Management, Virtualization for Data-Centre Automation.	**08 Hours**	**15%**
3	**Cloud Architectures** Workload Distribution Architecture, Resource Pooling Architecture, Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture, Hypervisor Clustering Architecture, Load Balanced Virtual Server Instances Architecture.	**08 Hours**	**15%**
4	**Edge Computing & its Applications** Edge computing purpose and definition, Benefits of Edge Computing, Different Types of Edge, Edge Deployment Modes, Edge computing hardware architectures (Gateway), Edge Computing Use-Cases, Edge Computing Marketplace.	**06 Hours**	**15%**
5	**Fog Computing & its Applications** Introduction to Fog Computing: Fog Computing, Characteristics, Application Scenarios, Issues and challenges. Fog Computing Architecture: Communication and Network Model, Programming Models, Fog Architecture for smart cities, healthcare and vehicles.	**06 Hours**	**15%**

6	**Container Technology & Tools** Understanding Basic Terms: Cgroups, Namespace, Layered File System etc., Understanding & Implementing Container, Virtual Machine vs Containers, Pros and Cons of Container Technology, Fundamentals of Docker, Docker networking and storage, Docker Compose, Introduction to Container Orchestration and Tool: Kubernets	**06 Hours**	**20%**
7	**Market Place of Advanced Computing Platforms** Study of Futuristic computing: Amazon Web Services, Microsoft Azure Services, Google Cloud Platform, Salesforce Enterprise Cloud Services.	**03 Hours**	**5%**

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Assess and examine advantages and disadvantages of cloud computing and virtualization technology.
CO2	Compose services in a distributed computing environment to achieve tasks relevant to a knowledge-based business or public service
CO3	Evaluate a set of business requirements to determine suitability for a cloud computing delivery model.
CO4	Explore the various cloud computing architectures and paradigms.
CO5	Deployment of cloud and identify security implications in cloud computing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	2	2	2	2	-	2	-	-	-	2	2	-
CO2	3	2	3	2	3	2	2	-	-	-	2	2	2	2
CO3	3	2	2	3	3	2	2	-	2	2	-	-	2	2
CO4	3	2	2	2	2	-	-	2	2	-	-	-	2	2
CO5	3	2	3	2	3	2	2	2	-	2	-	2	2	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial(High); If there is no correlation, put "-"

Recommended Study Material:

❖ **Text Books:**

1. Thomas Erl, Zaigham Mahmood,and Ricardo Puttini, “Cloud Computing Concepts, Technology & Architecture”, Prentice Hall
2. Kai Hwang, Geoffrey C.,”Distributed and Cloud Computing”, Morgan Kaufmann is an imprint of Elsevier
3. Nvin Sabharwal, Ravi Shankar “Apache CloudStack Cloud Computing” PACKT Publishing
4. Fog and Edge Computing: Principles and Paradigms by Rajkumar Buyya, Satish Narayana Srirama, wiley publication, 2019, ISBN: 9781119524984

❖ **Reference Books:**

1. Ravi Shankar, Navin Sabharwa “Cloud Computing First Steps: Cloud Computing for Beginners” Create Space Independent Publishing Platform
2. Rajkumar Buyya, James Broberg, Andrzej Goscinski “Cloud Computing: Principles and Paradigms” Wiley
3. Judith Hurwitz, Robin Bloor “Cloud Computing For Dummies” , for Dummies
4. IoT and Edge Computing for Architects - Second Edition, by Perry Lea, Publisher: Packt Publishing, 2020, ISBN: 9781839214806
5. David Jensen, “Beginning Azure IoT Edge Computing: Extending the Cloud to the Intelligent Edge, MICROSOFT AZURE

❖ **Web material:**

1. http://www.console.cloud.google.com
2. http://www.qwicklabscom
3. http://codelabs.developers.google.com
4. http://www.docker.com

❖ **Software/Platform:**

1. NetBeans
2. Eclipse
3. .NET
4. Google Cloud Platform
5. Amazon Web services
6. Microsoft Azure Platform

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS473: Blockchain Technology & Applications

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	4	2	6	5
Marks	100	50	150	

Outline of the Course:

Sr No.	Title of the unit	Minimum number of Hours
1	Backbones of Blockchain: Cryptography and Hashing	07
2	An Introduction to Blockchain Technology	06
3	Block-Chain Consensus Mechanisms	12
4	Introduction to Cryptocurrency: BitCoin	12
5	Blockchain hyperledger	05
6	Solidity Programming and Decentralized Application	08
7	Blockchain in Real World	10

Total Hours (Theory): 60

Total Hours (Lab): 30

Total Hours: 80

Detailed Syllabus:

1.	**Backbones of Blockchain: Cryptography and Hashing**	**07 hours**	**12%**
	Symmetric key cryptography, Public key cryptography, Digital Signature, Cryptographically Secured Hash Functions, Cryptographically Secured Chain of Blocks, Merkle Trees.		
2.	**An Introduction to Blockchain Technology**	**06 hours**	**10%**
	Evaluation of Blockchain Technology, Distributed Systems, The History of Blockchain and Bitcoin, Types of Block-Chain		

3	**Block-Chain Consensus Mechanisms**	**12 hours**	**20%**
	Practical Byzantine fault tolerance algorithm, Proof of Work, Proof of Stake, Proof of Authority, Proof of Elapsed time.		
4	**Introduction to Cryptocurrency: BitCoin**	**12 hours**	**20%**
	Digital keys and addresses, Transactions, Mining, The Bitcoinnetwork, Wallets, Bitcoin payments		
5	**Blockchain hyperledger**	**5 hours**	**8%**
	Fabric architecture, implementation, networking, fabric transactions, demonstration, smart contracts		
6	**Solidity Programming and Decentralized Application**	**08 hours**	**13%**
	Using the Remix IDE, Data Types and Functions, Creating Inline Assembly Functions, Mappings, Modifiers, Structs, and More, Generating ERC-20 Tokens, Extending Token Security (ERC-223),Deploying the ERC-20 TokenContract, DAPP using Ehtereum, DAPP using Hyperledger.		
7	**Blockchain in Real World**	**10 hours**	**17%**
	Blockchain applications, egovernance, smart cities, smart industries, use cases, trends on blockchains, serverless blocks, scalability issues, blockchain on clouds, Blockchain in IOT.		

Course Outcome (COs):
At the end of the course, the students will be able to

CO1	Understand the Basic Cryptography behind the Blockchain Technology and Bitcoin.
CO2	Define the structure of a Blockchain and classify why and when it is better than a simple distributed database.
CO3	Analyse the consensus mechanisms in a Blockchain Technology and critically assess its applicability in Blockchain based application.
CO4	Analyse to what extent smart and self-executing contracts can benefit automation, governance and transparent environment.
CO5	Design decentralized distributed application and measure the performance of Blockchain against centralized system.
CO6	Attain awareness of the new challenges that exist in monetizing businesses around Blockchain.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	-	-	1	-	-	2	1	-	-	-	2	-
CO2	-	2	-	-	-	2	1	1	1	-	2	-	1	2
CO3	1	3	2	1	-	2	-	-	2	-	1	2	3	-
CO4	1	1	2	2	3	2	2	2	2	-	3	2	3	1
CO5	1	3	3	2	3	2	2	2	3	2	3	2	3	2
CO6	1	3	1	3	-	2	2	-	3	1	3	2	3	2

Recommended Study Material:

❖ **Text Books:**

1. Imran Bashir , Mastering Blockchain ,Packt Second Edition, 2018.
2. Andreas M. Antonopoulos, Mastering Bitcoin, O'Reilly, Second Edition

❖ **Reference Books:**

1. Samanyu, Blockchain Developer's Guide, Packt, 2018.
2. Mark Watney, Blockchain for Beginners: The Complete Step by Step Guideto Understanding Blockchain Technology, July, 2017.
3. DonTapscott, Blockchain Revolution: How the Technology Behind Bitcoin Is Changing Money, Business, and the World, Hardcover, May 2016.
4. Xun (Brian) Wu, Hyperledger Cookbook, Packt.
5. Mayukh Mukhopadhyay, Ethereum Smart Contract Development, Packt,2018.

❖ **Web Materials:**

https://medium.com/topic/blockchain

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS474: Image Processing and Computer Vision

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	_	6	**5**
Marks	100	50	_	150	

Outline of the course

Sr. No.	Title of the unit	Minimum number of hours
1.	Digital Image Processing Fundamentals	02
2.	Segmentation of Grey level images	03
3.	Detection of edges and lines in 2D images	04
4.	Images Enhancement	05
5.	Introduction to Computer Vision	05
6.	Feature detection and matching	07
7.	Object Detection and Recognition	07
8.	Deep learning for Computer Vision	12

Total hours (Theory): 45
Total hours (Lab): 30
Total hours: 75

Detailed Syllabus:

1.	**Digital Image Processing Fundamentals**	**02 Hour**	**04 %**
	A simple image model, Sampling and Quantization, Imaging Geometry, Digital Geometry, Image Acquisition Systems, Different types of digital images		
2.	**Segmentation of Grey level images**	**03 Hour**	**07 %**
	Histogram of grey level images, Optimal thresholding using Bayesian classification, multilevel thresholding, Segmentation of grey level images, Water shade algorithm for segmenting grey level image.		
3.	**Detection of edges and lines in 2D images**	**04 Hours**	**07 %**
	First order and second order edge operators, multi-scale edge detection, Canny's edge detection algorithm, Hough transform for detecting lines and curves, edge linking		

4.	**Images Enhancement**	**05 Hours**	**11 %**
	Point processing, Spatial Filtering, Frequency domain filtering, multi-spectral image enhancement, image restoration.		
5.	**Introduction to Computer Vision**	**05 Hours**	**13 %**
	Introduction and Challenges in Computer Vision, Applications in real world, Geometric primitives, 2D and 3D transformations, Orthographic & Perspective Projection		
6.	**Feature detection and matching**	**07 Hours**	**15 %**
	Harris Corners, Invariant feature point detector -SIFT, SURF, RANSAC for point matching, Edge detection - LOG, DOG, Canny, Scale-Space Analysis - Image Pyramids and Gaussian derivative filter, Line, circle & ellipse detectors (Hough Transform)		
7.	**Object Detection and Recognition**	**07 Hours**	**16 %**
7.1	Machine Learning and Pattern Recognition in computer vision, classification models, Dimensionality Reduction (Principle component analysis), Face detection with sliding window – Haar-features, Viola Jones method and Adaboost training algorithm, People detection with sliding window, SVM, Bag of visual words		
8.	**Deep learning for Computer Vision**	**12 Hours**	**27 %**
	NeuralNetworks Fundamentals, Past issues with deep networks, Convolutional Neural Networks (CNNs), training of CNNs, representation and transfer learning, CNNs in classification and recognition task.		

Course Outcome (CO)

CO1	Students will be able to understand the fundamentals of Image Processing
CO2	Students will learn different types of image segmentation techniques, edge detection and line detection
CO3	Students will be able to perform the image enhancement techniques
CO4	Students will be able to understand the basics of Computer Vision
CO5	Students will be able to perform feature detection, object detection and recognition
CO6	Students will be able to apply the Deep Learning techniques for different computer vision application

Co-Po Mapping Matrix (Table)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	1	-	3	1	2	-	3	1	3	-	1	2	2	-	-
CO2	-	2	1	-	-	2	1	-	1	2	-	-	-	2	1
CO3	1	-	-	2	1	-	-	2	-	-	2	1	-	-	-

CO4	-	1	-	1	1	1	-	1	-	1	1	1	3	1	-
CO5	2	1	-	-	-	1	-	-	-	-	-	-	-	-	-
CO6	-	-	2	-	3	-	2	-	2	-	-	3	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)
If there is no correlation, put "-"

Recommended Study Material:

❖ **Text Books:**

1. Digital Image Processing, R.C. Gonzalez, R.E Woods, Pearson Education
2. Computer Vision: A Modern Approach, D. A. Forsyth, J. Ponce, Prentice Hall

❖ **Reference Books:**

1. Digital Image Processing and Computer Vision, R. J. Schalkoff, John Wiley & Sons Australia
2. Computer Vision, L. Shapiro, G. Stockman, Prentice-Hall
3. Introductory Techniques for 3D Computer Vision, E. Trucco, A. Verri, Prentice Hall

CS475: SOFTWARE DEFINED NETWORKS

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours	4	2	-	6	5
Marks	100	50	-	150	

B. Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Elements of Modern Networking	12
2.	SDN: Background and Motivation	12
3.	SDN Data Plane and OpenFlow	08
4.	SDN Control Plane & SDN Application Plane	12
5.	SDN via APIs	08
6.	Data centers definition	08

Total hours (Theory): 60

Total hours (Lab): 30

Total hours: 90

C. Detailed Syllabus:

1.	**Elements of Modern Networking**	**12 Hours**	**20%**
	The Networking Ecosystem, Example Network Architectures, Ethernet, Wi-Fi,4G/5G Cellular, Network Convergence, Unified Communications, Types of Network and Internet Traffic, Demand: Big Data, Cloud Computing, and Mobile Traffic, Requirements: QoS and QoE, Routing, Congestion Control, SDN and NFV, Modern Networking Elements		
2.	**SDN: Background and Motivation**	**12 Hours**	**20%**
	Evolving Network Requirements, The SDN Approach, SDN- and NFV-Related Standards		
3.	**SDN Data Plane and OpenFlow**	**08 Hours**	**13%**
	SDN Data Plane, OpenFlow Logical Network Device, OpenFlow Protocol		

4.	**SDN Control Plane & SDN Application Plane**	**12 Hours**	**20%**
	SDN Control Plane Architecture, ITU-T Model, OpenDaylight, REST, Cooperation and Coordination Among Controllers, SDN Application Plane Architecture, Network Services Abstraction Layer, Traffic Engineering, Measurement and Monitoring, Security, Data Center Networking, Mobilityand Wireless, Information-Centric Networking		
5.	**SDN via APIs**	**08 Hours**	**14%**
	SDN via APIS, SDN via Hypervisor-Based Overlays, SDN via Opening Up the devices, Network function virtualization, Alternative Overlap And Ranking		
6.	**Data centers definition**	**08 Hours**	**13%**
	Data centres definition, Data centres demand, tunnelling technologies for Data centres Pathtechnologies in data centres, Ethernet fabrics in Data centres, SDN use case in Data centres.		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Measure and Analyse different network parameters.
CO2	Understand working of application layer protocols.
CO3	Understand and Analyse transport layer services and its impacts on data rate.
CO4	Understand basic functionality of network layer devices.
CO5	Understand functionality of multiple access protocol.
CO6	Analyse traditional network and get familiar with Software defined networking.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	1	3	1	-	-	-	-	-	-	-	2	-
CO2	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO3	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO4	3	3	1	3	1	-	-	-	-	-	-	-	1	-
CO5	3	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	-	-	-	-	3	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial(High)Ifthere is no correlation, put “-”

Recommended Study Material:

❖ **Text Books:**

1. William Stallings, Florence Agboma, Sofiene Jelassi “Foundations of Modern Networking, SDN, NFV, QoE, IoT, and Cloud”; Pearson Publisher, ISBN-13: 978-0-13-417539-3
2. Behrouz A. Forouzan, “TCP/IP Protocol Suite.”, Fourth Reprint, 2003;Tata McGraw Hill ISBN: 0-07-049551-3

❖ **Reference Books:**

1. Douglas E. Comer and David L. Stevens, “Internetworking with TCP/IP Volume-2,Design, Implementation and Internals ”, Prentice Hall

❖ **Web Materials:**

1. https://www.sdxcentral.com/
2. https://sdn.ieee.org/standardization
3. https://trac.ietf.org/trac/irtf/wiki/sdnrg
4. https://www.opennetworking.org/sdn-resources/openflow
5. https://www.opendaylight.org/
6. https://www.opennetworking.org

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS452: SOFTWARE GROUP POJECT - V

Credit and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	2	2	1
Marks	0	100	100	

Outline of the Course:

- Student at the beginning of a semester may be advised byhis/her supervisor (s) for recommended courses.
- Students will work together in a team (at most ***three***) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
  a. Project Synopsis
  b. Software Requirement Specification
  c. SPMP
  d. Final Project Report
  e. Project Setup file with Source code
  f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total Hours (Theory): 00
Total Hours (Lab):30
Total Hours: 30

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify problems present in society by surveying variety of domains and convert in project definition.
CO2	Explore new ideas and techniques to solve it. Create, select and apply appropriate techniques, resources, modern engineering and IT tools to solve problem.
CO3	Correlate knowledge of different subjects and apply theoretical knowledge to implement project for identified problem.
CO4	Apply ethical principles and commit to responsibilities and norms of theproject.
CO5	Writetechnical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO6	Apply engineering and management principles to achieve project goal.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	-	-	-	-	-	-
CO2	-	-	3	-	-	-	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO4	-	-	-	-	-	-	-	3	2	-	-	-	-	-
CO5	-	-	1	-	3	2	-	-	-	3	-	-	-	-
CO6	3	2	2	1	2	1	-	-	2	3	-	1	-	-

Enter correlation levels 1, 2 or 3 as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial
(High)If there is no correlation, put "-"

Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic

❖ **Web Materials:**

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. http://spie.org/x576.xml

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS446: SUMMER INTERNSHIP- II

Credit and Hours:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours/week	90	-	-	90	3
Marks	150	-	-	150	

Outline of the Course:

1.	**Instructional Method and Pedagogy**
	• Summer internship shall be at least 90 hours during the summer vacationonly. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University. • Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be

	compensated later.
	• The student shall submit two documents to the coordinator for the evaluation of the summer internship:
	  ◦ Summer Internship Report
	  ◦ Summer Internship Assessment Form
	• Upon the completion of summer internship, a hard copy of “Summer Internship Report” must be submitted through the presentation to the coordinator by the first dayof the new term.
	• The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
	• If the evaluation ofthe report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

2.	**Format of Summer Internship Report**
	The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed.An electronic copyof the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: • Cover Page • Page of Approval and Grading • Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. • TABLE OF CONTENTS (with the corresponding page numbers)

- LIST OF FIGURES AND TABLES (with the corresponding page numbers)
- DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
    - Location and spread of the company
    - Number of employees, engineers, technicians, administrators in the company
    - Divisions of the company
    - Your group and division
    - Administrative tree (if available)
    - Main functions of the company
    - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main bodyof the report), you can place them here. These could be schematics, algorithms, drawings, etc. Ifthe document is a datasheet and it

	can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don't have to print it and waste tons of paper.	
	Total hours (Project) :	90
	Total hours :	90

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact ofthe internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- **Reference book:**
  1. Books, Magazines & Journals of related topics
- **Web material:**
  1. www.ieeexplore.ieee.org
  2. www.sciencedirect.com
  3. www.elsevier.com
  4. http://spie.org/x576.xml
- **Software**
  1. ASP.NET
  2. PYTHON/MATLAB
  3. PHP
  4. ANDROID/IOS
  5. FLUTTER
  6. NODE/REACT NATIVE

B.Tech. (Computer Science & Engineering) Programme

SYLLABI

(Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF TECHNOLOGY & ENGINEERING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CS453: SOFTWARE PROJECT MAJOR

Credit Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	0	36	36	18
Marks	0	600(250+350)	600	

Outline of the Course:

- SoftwareProject includes course work on a specialized Subject or aSeminar.
- The course work shall be related to the area of his/her project research work.
- Students have to take 3 months training to the other software industry as the project work.
- The major project work provides students an opportunity to do something on their ownand under the supervision of internal guide as well as guide from industry.
- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Project will be evaluated at least thrice during the semester by internal guide of the project and final submission at the end ofthe semester as a part ofcontinuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as solution of particular problem by applying principles of Software Engineering.
- A student has to produce some useful outcome by conducting experiments or project work.
- Student can learn all aspects & functionalityof specialized software from the industry.
- Students have to submit SRS, SPMP, Design documents, Code and Test Cases in formof Project report.

Instructional Method and Pedagogy:

- Students can select the topic based on the industry requirement.
- Students can select thetopic based onthe research areasof available supervisor/guide.
- Each student has to prepare a synopsis, SRS, SPMP, Design documents and Final Reportin prescribed format only. The report typed on A4 sized sheets and bound should be submitted after approval bythe guide and endorsement of the Head of Department.

- Performance of students at the Project will be assessed by the department faculty. The supervisor/guide based on the quality of work and preparation and understanding of thecandidate shall do an assessment of the project.

- Following format for documentation for the project be followed:
  A. Forwarding Page
  1. Title of the project
  2. Objectives
  3. Definitions of Key Term Approach to Problem solving Limitations, if any
  4. Output Generated
  5. Details of Hardware Performed
  6. Details of Software Tools used
  7. Implementation issues (clearly defining the area ofapplication)
  8. Miscellaneous
  9. Signature of candidate & date

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify and justify/analyse the requirements of the projects with enhancement of the required tools and technology by individual and team.
CO2	Solve challenging projects for commercial, societal and environment benefit.
CO3	Applythe engineering knowledge to full fill the requirements of the projects pertaining to anydiscipline.
CO4	Explain the importance of planning, documentation, punctualityand work ethics.
CO5	Document the work which is carried out in proper format with industrystandards.
CO6	Showcase the soft skill.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	3	-	1	-	3	1	1	1	3	1
CO2	2	-	3	-	-	3	2	-	-	-	-	-	2	2
CO3	3	2	1	1	1	-	-	-	1	-	2	1	3	1
CO4	3	1	2	-	-	-	-	3	-	-	1	2	2	3
CO5	3	-	2	2	-	-	-	-	-	2	-	3	1	1
CO6	1	-	1	1	-	-	-	3	1	3	1	1	1	3

Enter correlation levels 1, 2 or 3 as defined below:
1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)If there is no correlation, put "-"

Recommended Chapters/sections

1. Microscope Summery
2. Details of candidate and supervisor along with certificate of
   - original work;
   - Assistance, if any;
3. Credits; Aims and Objectives
4. Approaches to Project and Time Frame
5. Project Design Description with appendices to cover

   Flow charts/Data Flow Diagram – Macro/Micro Level

   Source code, If any

   Hardware platform

   Software Tools

   Security Measures

   Quality Assurance

   Audit ability
6. Test Date and Result

Recommended Study Material:

1. Reading Materials, web materials, Project reports with full citations
2. Books, magazines & Journals of related topics
3. Various software tools and programming languages compiler related to topic

❖ **Web Materials:**

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. http://spie.org/x576.xml